COVERAGE-INVERSION: A CALIBRATION-TRANSPARENT ORACLE PRIMITIVE FOR TOKENIZED RWAS

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Abstract

Tokenised real-world assets on public blockchains trade continuously, but their underlying price-discovery venues do not, leaving roughly two-thirds of wall-clock time during which on-chain protocols must value collateral against a closed reference market. We invert the conventional point-plus-confidence oracle interface so that target coverage $\tau \in (0, 1)$ is a consumer request and the served claimed quantile q_{served} is a per-read receipt, and we build the empirical calibration surface $S^f(s, r, q)$ that maps a base forecaster f, an asset s, and a regime label $r = \rho(\mathcal{F}_t)$ to its historical realised coverage at each grid quantile. The resulting primitive carries a three-property contract: auditability of the surface against public data (P_1), conditional empirical coverage of the served band at τ (P_2), and per-regime serving efficiency over a candidate forecaster set $\mathcal{F} = \{\texttt{F0_stale}, \texttt{F1_emp_regime}\}$ (P_3). We evaluate on 5,986 weekend prediction-window rows across 10 US-listed tickers (638 distinct weekends, with HOOD partial post-2021) from 2014-01-17 through 2026-04-17, with a temporally disjoint 2023+ held-out slice of 1,720 rows over 172 weekends; at the consumer-facing target $\tau = 0.95$ the deployed serving configuration — calibration surface frozen on the pre-2023 calibration set and a per-anchor empirical buffer schedule deployment-tuned on the held-out slice — realises 0.950 coverage with Kupiec $p_{uc} = 1.000$ and Christoffersen $p_{ind} = 0.485$; a six-split expanding-window walk-forward ratifies the deployed buffer at $\tau = 0.95$ as the cross-split mean (0.020 deployed vs 0.019 mean, $\sigma = 0.017$), so the operating result is deployment-calibrated rather than purely held-out end-to-end. Berkowitz and DQ still reject, which narrows the empirical claim to per-anchor calibration rather than full-distribution calibration and, more importantly, identifies a concrete V2 research agenda: rolling live rebuilds, richer on-chain signals, and finer regime structure. An eight-rung forecaster ladder plus five-cell serving-layer matrix isolate the load-bearing components with block-bootstrap confidence intervals. The reference implementation is open-sourced as a Python research stack and a Rust + Anchor production stack with byte-for-byte parity (75/75 cases) on a Solana on-chain publisher. We do not claim point-accuracy state-of-the-art, integrity against adversarial feeds, or an optimal liquidation-policy default; those concerns sit in adjacent companion papers.

1 Introduction

Tokenized equities and other real-world assets (RWAs) on public blockchains trade 24 hours a day, 7 days a week. Their underlying price-discovery venues do not. A US equity listed on NASDAQ has a regular trading session of 6.5 hours per weekday, or roughly 32% of wall-clock time. In the other 68% — nights, weekends, halts, and extended holiday closures — the price continues to change: earnings are released, macro prints land, geopolitical events clear, and overseas sessions trade correlated instruments. But the venues that establish reference prices are closed.

This gap between continuous on-chain demand and discontinuous off-chain price discovery is a structural feature of the tokenized-RWA architecture. It is also where the great majority of adverse selection, undercollateralized liquidations, and protocol-level loss events have historically concentrated. Any decentralized-finance protocol that accepts tokenized

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RWAs as collateral or as inputs to an automated market maker must resolve a concrete question at every block: *at this on-chain moment, with the underlying venue closed, what is the fair price — and how uncertain is it?*

The same question recurs across the broader RWA wave that is moving on-chain in 2025–2026: tokenized equities (xStocks on Solana, Backed-issued tokens across multiple chains), tokenized treasuries (BlackRock BUIDL, Ondo OUSG; collectively > $5B AUM as of submission), tokenized commodities (Paxos PAXG, forthcoming silver and other metals), and the next wave of tokenized credit and FX. Every asset class with a continuous off-hours information set — futures, volatility indices, spreads, overseas-session prices — inherits the same closed-market-pricing problem and would benefit from the same calibration-transparent risk-reporting primitive. We argue that this primitive is the *missing infrastructure layer* for responsible institutional-scale RWA onboarding, in the same way that block explorers and indexers are missing infrastructure that the ecosystem depends on but does not directly produce. The methodology fits asset classes with continuous off-hours data; it does not fit instruments without one (e.g., illiquid commodities or pure NAV-update real estate). Section 10.5 enumerates the generalization map and §10.6 the asset classes that fall outside the methodology's domain.

1.1 How existing oracles answer, and what they leave unanswered

Three classes of RWA price oracle are deployed at scale today. Each resolves the *what* of the question above, and each silences the *how uncertain*.

Stale-hold. Chainlink Data Streams [Labs, 2024] for tokenized US equities exposes a `price` field that holds the venue's last observed trade and flags it as closed until the venue reopens — `marketStatus = 1` ("Closed") on the v10 Tokenized Asset schema, whose code set is the coarse `{0 Unknown, 1 Closed, 2 Open}`, and `marketStatus = 5` ("Closed/weekend") on the v11 RWA Advanced schema, whose finer code set distinguishes pre-market, regular, post-market, overnight, and weekend states. In either schema the published value is held static across the closed interval; the uncertainty signal is binary (stale or live), so a consumer cannot distinguish a two-hour halt from a three-day weekend from a long Presidents' Day weekend ending an earnings week, even though the *expected* drift across those three intervals differs by more than an order of magnitude. The v10 schema additionally exposes a continuously-updating `tokenizedPrice` field — empirically observed to drift sub-second across weekends — but with limited public methodology disclosure (Chainlink's tokenized-asset documentation describes the field qualitatively as a CEX-aggregated mark; the aggregation rule, contributor set, and any smoothing are not specified at a level that supports reproducing or auditing a coverage claim); that field falls under the third archetype below. The separately-deployed v11 schema for the same underlyings carries `bid/ask/mid/last_traded_price` fields that are placeholder-derived or frozen during the same weekend window, behaving as the stale-hold archetype on every published field.

Quote-dispersion confidence intervals. Pyth Network [Network, 2021, 2022] publishes an aggregate of prices from permissioned publishers together with a confidence interval σ derived from the dispersion of those quotes. During market hours this σ is an aggregation diagnostic — it measures how much publishers disagree. During closed-market hours, when the contributor set thins, σ becomes noisy; critically, it is never asserted to be the standard deviation of the P_t distribution *conditional on* time-of-week, elapsed interval, or realised volatility regime. Protocols that read Pyth's σ as a probability-of-coverage statement are extrapolating.

Undisclosed methodology. RedStone Live [RedStone, 2026] and several smaller providers offer 24/7 equity feeds whose closed-market value is computed by methods that are not documented to a level that supports reproducing or auditing a coverage claim. RedStone's coverage in this category is currently for the *underlier ticker* (e.g., SPY) rather than the on-chain SPL xStock token (SPYx), so a Solana protocol holding SPYx as collateral reads a related but not contractually-identical instrument; Cong et al. [Cong et al., 2025] document that off-hour returns on tokenized stocks anticipate rather than amplify subsequent Monday opens, so the on-chain instrument is informationally distinct from its underlier in exactly the closed-market window the oracle covers. Chainlink Data Streams' v10 `tokenizedPrice` field shares the methodology-disclosure archetype on the same empirical observation: continuous off-hours updating with only qualitative public disclosure of the aggregation rule. A consumer cannot reconstruct the value or audit whether the served band corresponds to any particular probability-of-coverage level. The feed is consumed on trust.

None of the three publishes a *calibration claim verifiable against public data at the aggregate feed level*: a statement of the form *"over N historical prediction windows, our 95% band contained the realised price N_{in} times, giving realised coverage N_{in}/N."* Pyth's documentation [Network, 2022] recommends that *individual publishers* calibrate their submitted confidence intervals to approximately 95% coverage, but this is a per-publisher self-attestation rather than a verifiable property of the aggregate served feed; downstream protocols read the aggregate. None of the three publishes the rolling dataset from which an aggregate-level calibration claim could be independently reproduced.

1.2 Coverage as a first-class API parameter

The design point of this paper is an RWA oracle whose *product contract* is a calibration statement. We invert the usual interface. Instead of publishing a single band and leaving its coverage level implicit, we take a consumer-specified target coverage $\tau \in (0, 1)$ as a request parameter and return the band that has *empirically* delivered coverage τ on a rolling historical sample, stratified by asset and pre-publish regime.

Concretely, for a base forecaster f producing bands $[L_t^f(s; q),\ U_t^f(s; q)]$ at a grid of claimed quantiles q, we build the empirical calibration surface

$$S^f(s, r, q) \;=\; \frac{1}{|\mathcal{T}_{s,r}|} \sum_{t \in \mathcal{T}_{s,r}} \mathbf{1}\big[\, L_t^f(s; q) \le P_t(s) \le U_t^f(s; q) \,\big]$$

over historical prediction windows $\mathcal{T}_{s,r}$ in each (symbol, regime) bucket. At serve time, we invert: the band delivered for a request (s, t, τ) is the one at the interpolated $q_{\text{served}} = (S^f)^{-1}(s, \rho(\mathcal{F}_t), \tau)$.

Two properties follow from this construction that are absent in the incumbent designs:

1. **Auditability.** S^f is a deterministic function of public data (prices, implied-vol indices, calendar flags), the choice of base forecaster f, and a documented regime labeler ρ. Any third party with access to those inputs can reconstruct S^f and verify that a served band corresponds to the published q_{served} on the claimed surface.
2. **Served-coverage receipts.** Every oracle read exposes the tuple $(q_{\text{served}},\ f,\ r)$ alongside the band. A protocol integrator can log this tuple, backtest the oracle against its own loss function, and challenge the provider if the realised rate drifts.

We are not the first to argue for calibration in financial forecasting — the framework traces to Christoffersen [Christoffersen, 1998] and has institutional instantiations in SR 11-7 [of Governors of the Federal Reserve System & Office of the Comptroller of the Currency, 2011] / SR 26-2 [Board of Governors of the Federal Reserve System, 2026] model-risk management guidance. What is new here is the application to a decentralised oracle with a consumer-facing SLA on realised coverage rather than on point-estimator accuracy.

1.3 What this paper shows

We evaluate the coverage-inversion primitive on a panel of 5,986 weekend prediction-window rows (638 weekends × 10 symbols, with HOOD partial post-2021) spanning 2014-01-17 through 2026-04-17: seven liquid US equities and ETFs (SPY, QQQ, AAPL, GOOGL, NVDA, TSLA, HOOD), one leveraged-gold ETF (GLD), one long-duration treasury ETF (TLT), and one Bitcoin-proxy equity (MSTR). Each prediction window runs from Friday 16:00 ET to Monday 09:30 ET, the contiguous non-trading interval that produces the largest expected drift and the most adverse oracle-failure cases.

Our base forecaster combines a factor-switchboard point estimate (Friday close scaled by a per-asset-class futures return: ES/NQ/GC/ZN/BTC) with a log-log regression on a per-symbol volatility index (VIX / GVZ / MOVE) for the conditional sigma, and constructs the band by empirical quantiles of the standardised residual. The forecaster is deliberately simple. We do not claim it is minimum-variance. The contribution of the paper is the *primitive* that surrounds any such forecaster — the calibration surface, its inversion, and the audit receipts — together with empirical evidence that this primitive delivers its coverage promise on held-out data.

The headline empirical claims are:

- At consumer target $\tau = 0.95$, the deployed serving configuration on a temporally disjoint 2023+ slice (1,720 rows, 172 weekends) realises **95.0%** coverage, with Kupiec unconditional-coverage test p-value of **1.000** (not rejected) and Christoffersen independence p-value of **0.485** (not rejected). The calibration surface is held-out (frozen on the pre-2023 calibration set; no parameter, threshold, or grid point is updated using OOS data); the per-anchor empirical buffer schedule of §4.3 is *deployment-tuned* on the same OOS slice. The strongest defensible characterisation of this number is therefore "deployment-calibrated operating result on a held-out surface", not "purely held-out end-to-end". A six-split expanding-window walk-forward (§9.4 / `reports/v1b_walkforward.md`) ratifies that the deployed buffer at $\tau = 0.95$ (0.020) lands at the cross-split mean (0.019, $\sigma = 0.017$), which is the empirical evidence that the deployment-tuned value is not idiosyncratically overfit to the 2023+ slice. Reporting Kupiec + Christoffersen on a temporally disjoint slice is the model-risk-management standard format that matches the Basel backtesting framework; we adopt it

because protocol integrators and their regulators already speak it. We use $\tau = 0.95$ as the paper's primary *oracle-validation* target; we do not claim in this paper that $\tau = 0.95$ is the optimal liquidation-policy operating point for any particular lending protocol. The deployed default for protocol consumption is $\tau = 0.85$, picked on protocol-EL grounds against a Kamino flat ±300 bps benchmark (forward-referenced to Paper 3 of the trilogy; see §10.4); at that target realised coverage is 0.855 with both tests passing.

- An ablation across eight forecaster variants with block-bootstrap 95% confidence intervals isolates the contribution of each component. The factor-switchboard point and empirical-quantile residual quantile together produce a **39.3% (CI [37.3%, 41.3%])** reduction in pooled band half-width versus a stale-Gaussian baseline. The log-log vol regression is regime-localised: it lifts high-volatility coverage by 10.4pp cumulatively at the cost of wider bands, explaining why the serving-layer hybrid policy switches to the stale-Gaussian band in that regime.
- The per-target buffer schedule on the serving path is the OOS coverage workhorse: replacing surface-inversion-only with a per-anchor empirical buffer (deployed values $\{0.68 \to 0.045, 0.85 \to 0.045, 0.95 \to 0.020, 0.99 \to 0.010\}$) closes the coverage gap by $+2.9$pp (CI [$+2.0$, $+3.8$]) at $\tau = 0.95$. The hybrid serving policy contributes essentially nothing to mean coverage on OOS data but is the *empirically preferred deployment choice* on the evaluated sample, weakly dominating F1-everywhere on both sharpness and the Christoffersen-independence margin (an evaluation diagnostic, not a decision-theoretic optimisation target): the hybrid simultaneously tightens bands (-2.6% vs F1-everywhere) and improves the Christoffersen margin ($p_{ind} = 0.485$ vs 0.221), with no observed trade-off.
- The same diagnostics that keep the paper honest also define the V2 roadmap. Berkowitz rejects, so the present system should be read as a per-anchor calibrated interval oracle rather than a full-distribution-calibrated density forecast; DQ rejects, so residual multi-lag conditional structure remains after the current three-state regime labeler and buffer schedule are applied. We treat those findings not as contradictions of the primitive but as evidence about where the next gains should come from: rolling live rebuilds, finer regime structure, and richer on-chain signals (§10).

These claims are about the oracle interface and its serving-time calibration behavior. They are not a proof that the same target τ or the same serving policy is the welfare-optimal liquidation rule for a lending protocol; that stronger decision-theoretic question requires an explicit model of portfolio weights, liquidation costs, and protocol semantics, and is deferred to future work.

1.4 Contributions and structure

This paper makes four contributions.

C1. Coverage inversion as an oracle primitive. We formalise (§3) the calibration-surface construction and its serving-time inversion, and position it as a consumer-facing SLA distinct from existing point-plus-CI interfaces.

C2. A 12-year empirical calibration surface on public data. We describe (§4, §5) a simple, reproducible forecaster and regime labeler whose calibration surface can be rebuilt by any third party from freely available data. Our artifact includes the persisted surface, the bounds table, and the serving-time Oracle code.

C3. Held-out-surface, deployment-calibrated Kupiec + Christoffersen validation. We report (§6) conditional-coverage diagnostics in the form required by institutional model-risk management, on a slice that is temporally disjoint from the surface's training window. The calibration surface is held-out; the per-anchor empirical buffer is deployment-tuned on the same slice and walk-forward-stable across six expanding-window splits (§6.4). DQ and Berkowitz disclosures (§6.4.1) narrow the empirical content to per-anchor calibration in the Kupiec + Christoffersen sense rather than full-distribution calibration.

C4. An ablation with bootstrap confidence intervals. We quantify (§7) which components of the forecaster and serving layer are load-bearing for the calibration claim, and which are disclosed for auditability but contribute no measurable signal at our sample size.

Section 2 surveys related work in oracle design, price discovery across venues, and calibrated / conformal prediction. Section 3 gives the formal problem statement. Section 4 describes the methodology. Section 5 describes the data and regime labeler. Section 6 reports calibration results. Section 7 presents the ablation study. Section 8 describes the serving-layer system (Rust runtime, Anchor on-chain publisher, parity verification). Section 9 enumerates limitations and the assumptions under which the coverage-inversion claim holds. Section 10 lists future work including a conformal upgrade and a systematic incumbent-oracle benchmark.

2 Related Work

Our contribution sits at the intersection of three literatures that have, to our knowledge, never been brought together in a deployed system: decentralised-oracle design, cross-venue price discovery, and the calibration / conformal-prediction framework for interval forecasts. A fourth literature — the institutional model-risk-management guidance that governs how regulated firms backtest interval forecasts — supplies the disclosure vocabulary we adopt. We survey each in turn, indicating in each paragraph the specific claim from §1 that the citation supports or the prior design we distinguish against.

2.1 Decentralised oracle designs

The deployed oracle landscape on public blockchains is organised around an *integrity* primitive: given a real-world observable, can the oracle deliver a faithful value on-chain without a trusted intermediary? Each of the dominant designs answers this question differently. Chainlink's decentralised-oracle-network architecture [Breidenbach et al., 2021] aggregates off-chain reports from a committee of node operators and delivers a signed report on-chain, with the Data Streams product [Labs, 2024] providing low-latency pull-based delivery and an explicit `marketStatus` flag on real-world-asset feeds. Pyth Network [Network, 2021, 2022] takes a first-party-publisher approach on Solana: each permissioned publisher submits a price and a confidence interval, and the aggregate rule assigns each publisher three votes (price, price ± confidence) before taking the median. Switchboard [Switchboard, 2023] generalises the aggregation layer into permissionless oracle queues with TEE-attestation, and RedStone's recent Live product [RedStone, 2026] blends institutional feeds with perpetual-market data during off-hours. Optimistic-oracle designs — UMA [Project, 2020] and Tellor [Tellor, 2023] — replace continuous aggregation with a dispute-and-stake mechanism, resolving observable values by economic challenge rather than by committee vote.

None of these designs publishes a *calibration claim* on the served value. Pyth's confidence interval is well-documented [Network, 2022] as an aggregation diagnostic — it measures publisher disagreement and is asserted at the publisher level to be approximately 95% coverage on the spot price, but the aggregate-level interval widens mechanically as contributor sets thin, which is precisely what happens during off-hours for real-world-asset feeds. Chainlink Data Streams' v10 `price` field for tokenized US-equity feeds holds the last observed venue trade and flags it as closed (`marketStatus = 1` on the v10 Tokenized Asset schema, whose code set is the coarse `{0 Unknown, 1 Closed, 2 Open}`; `marketStatus = 5` on the separately-deployed v11 RWA Advanced schema, whose finer code set is `{0, 1 Pre-market, 2 Regular, 3 Post-market, 4 Overnight, 5 Closed/weekend}`) until the venue reopens [Labs, 2024]; in either case the served value carries no probability-of-coverage statement, and a consumer cannot distinguish a two-hour halt from a three-day weekend. The same v10 schema's `tokenizedPrice` field updates continuously across the closed interval, but the public methodology disclosure is qualitative (a CEX-aggregated mark with the aggregation rule, contributor set, and any smoothing not specified at a level that supports auditing a coverage claim) — empirically the field drifts through several hundred basis points across a single weekend in our V5 tape — and so shares the auditability gap of RedStone's product rather than constituting an alternative archetype. The separately-deployed v11 schema for the same underlyings exposes `bid/ask/mid` fields that are synthetic placeholders during weekend windows and a `last_traded_price` that is frozen at Friday's close, so on every published v11 field the served value is in the stale-hold class. RedStone's off-hours methodology is described qualitatively in the Live launch post — institutional feeds during market hours, perpetual-market data during off-hours — but not in reproducible form [RedStone, 2026]; the public REST gateway exposes no confidence, dispersion, sample-count, or band field on any served record, and the served instrument is the underlier (e.g., SPY) rather than the on-chain SPL xStock token (SPYx), which is informationally distinct in the closed-market window per [Cong et al., 2025]. The full empirical scoping of a matched-window RedStone comparison — coverage gaps, retention cap, and what is unverifiable about the paid Live tier — is in §9.8.1. Optimistic oracles [Project, 2020, Tellor, 2023] guarantee *eventual* correctness under economic security but do not define a distributional statement conditional on time-of-week or realised volatility. Our claim in §1 that no incumbent oracle publishes a calibration statement verifiable against public data is, on the evidence of these primary sources, not rhetorical: it is the direct reading of the deployed systems.

The academic literature on decentralised oracles has similarly concentrated on integrity. Angeris and Chitra [Angeris and Chitra, 2020] prove conditions under which constant-function market makers are incentive-compatible as price oracles for on-chain assets, focusing on arbitrage frictions rather than on the distributional properties of the reported series. The SoK by Eskandari et al. [Eskandari et al., 2021] catalogues oracle-manipulation attacks — flash-loan, liquidity, centralised-feed failures — and surveys mitigations; the framing throughout is adversarial (*did the attacker corrupt the value?*) rather than statistical (*over N windows, did the served band contain the truth at the claimed rate?*). The BIS bulletin on the oracle problem [Duley et al., 2023] frames the tension as a decentralisation-versus-efficiency trade-off and is sceptical of real-world-asset use cases specifically because introducing trusted parties back into the

system undermines the trustlessness that is the oracle's whole point. On-chain consumers operate under adversarial ordering [Daian et al., 2020], which amplifies the cost of any systematic band miscalibration.

Our contribution is orthogonal to this prior work. The integrity primitive asks whether the oracle reports $\hat{P}_t \approx P_t^{\text{true}}$; the calibration primitive asks, given the oracle's own band $[L_t, U_t]$, whether the served band contains P_t^{true} at the claimed rate on rolling historical evidence. An oracle can be integrity-perfect and calibration-meaningless: Chainlink Data Streams faithfully reports Friday's 16:00 close on Sunday at 14:00, but that value is not asserted to be a 50%- or 95%-coverage estimate of Monday's opening price. We take this observation as the design point, and invert the interface so that τ is a request parameter and q_{served} is the receipt.

2.2 Cross-venue price discovery

The base forecaster that underlies our calibration surface is not methodologically novel; its point estimate is a factor-switchboard scaling of Friday's close by a weekend futures return, and the conditional-sigma uses a log-log regression on a regime-matched volatility index. The choice of futures as the leading venue is a direct application of Hasbrouck's information-share methodology [Hasbrouck, 1995], which decomposes the efficient-price innovation variance across venues of a cross-listed security via a vector-error-correction model. For US equity indices, this methodology [Hasbrouck, 2003] yields the empirical result that most of the price discovery in the S&P 500 and Nasdaq-100 occurs in the E-mini futures market — the quantitative statement we rely on when we scale SPY and QQQ Friday closes by ES and NQ weekend returns respectively. The Gonzalo–Granger component-share decomposition [Gonzalo and Granger, 1995] is the conventional complement. Earlier foundational results [Stoll and Whaley, 1990, MacKinlay and Ramaswamy, 1988] established the lead-lag in index-futures markets on 5-minute intraday scales and documented mispricing-adjustment dynamics of 15–40 minutes, grounding Hasbrouck's cleaner identification. None of this is ours to claim; our contribution is the primitive that *wraps around* any such forecaster, not the forecaster itself.

The closed-market interval we predict across is specifically the Friday 16:00 ET to Monday 09:30 ET weekend window, which is chosen because it is the longest contiguous non-trading interval in the US equity calendar and, by classical microstructure results, the largest expected-drift window. French [French, 1980] documents the Monday-effect — significantly negative mean Monday returns across 1953–1977 — and establishes the weekend as a distinct return-generating regime. Barclay and Hendershott [Barclay and Hendershott, 2003] show that after-hours price discovery, while less efficient per unit time than regular-hours discovery, is non-trivial: information continues to arrive, so a stale-hold feed is systematically miscalibrated with respect to the true P_t distribution. Their companion paper [Barclay and Hendershott, 2004] shows that after-hours effective spreads are 3–4× regular-hours spreads, reflecting greater adverse selection — which is exactly the window in which a protocol holding an RWA as collateral is most exposed to oracle-driven loss events. For ETFs specifically, Madhavan and Sobczyk [Madhavan and Sobczyk, 2016] characterise creation/redemption-driven price discovery and validate that our ETF-class instruments (SPY, QQQ, GLD, TLT) can reasonably be treated within a unified price-discovery frame.

A newer thread of this literature addresses the structural migration that motivates our work. Makarov and Schoar [Makarov and Schoar, 2020] document recurring cross-venue arbitrage in cryptocurrency markets and the role of capital controls in sustaining price deviations. Lehar and Parlour [Lehar and Parlour, 2025] show empirically that decentralised exchanges on Ethereum do not exhibit long-lived arbitrage relative to centralised benchmarks, confirming that on-chain prices carry genuine price-discovery content during venue-active hours. Closing the loop, Cong et al. [Cong et al., 2025] provide the first large-scale empirical study of tokenised stocks and find that token prices track underlying stocks closely in regular hours but that *off-hour* token returns anticipate — rather than amplify — Monday openings. Their result is the direct empirical motivation for our primitive: weekend price discovery for tokenised equities is actively migrating to crypto-native venues, so a calibrated fair-value band conditioned on the on-chain continuous clock, rather than a stale-hold of Friday's close, becomes a first-class protocol need.

2.3 Calibrated and conformal prediction

The evaluation framework we adopt in §6 is not new. Kupiec [Kupiec, 1995] proposed the unconditional-coverage likelihood-ratio test for VaR systems: given a nominal level α and N out-of-sample observations, the test asks whether the realised hit frequency is statistically consistent with α. Christoffersen [Christoffersen, 1998] generalised this to conditional coverage, decomposing the evaluation into Kupiec's unconditional test and an independence test on the hit sequence, and combining them in a joint likelihood ratio. The p_{UC} and p_{ind} columns we report in §6 are exactly these two tests; the stated reason for using this pair (rather than the joint statistic alone) is that they correspond in banking practice to the Basel traffic-light framework [on Banking Supervision, 1996], which graduates regulatory responses by exception count over a rolling window. The density-forecast generalisation — Diebold, Gunther, and Tay [Diebold et al., 1998] — evaluates whether the PIT series is i.i.d. uniform, and Berkowitz [Berkowitz, 2001] refines this to a

small-sample-robust joint LR test on the inverse-normal-transformed PITs. Our surface-based coverage check at fixed τ is, in effect, a marginal slice of the PIT framework; we adopt it because the protocol-consumer SLA is per-τ, not distribution-wide.

The modern statement of *what* a good forecast interval is comes from Gneiting, Balabdaoui, and Raftery [Gneiting et al., 2007]: maximise sharpness subject to calibration. The paradigm — calibration is a necessary property of the joint distribution of forecasts and realisations, and sharpness measures the concentration of the forecast — is the framing under which our ablation (§7) is organised: each component of the forecaster either contributes to coverage (calibration) or to bandwidth reduction (sharpness), and we measure each with block-bootstrap confidence intervals. The Gneiting–Katzfuss survey [Gneiting and Katzfuss, 2014] updates this framework for the prediction-space setting. A recent refinement extends the framework specifically to the *tail* of the forecast distribution: Allen, Koh, Segers, and Ziegel [Allen et al., 2025] formalise *tail calibration* as a property distinct from full-distribution calibration, with diagnostic tools tied to peaks-over-threshold extreme value theory. This distinction is directly relevant to our setting — a coverage-inversion primitive that is calibrated on average can still be miscalibrated in the tail, which is the failure mode our $\tau = 0.99$ ceiling (§9.1) describes — and we treat their framework as the modern parent literature for the high-τ behaviour we report. The entire tradition traces to Brier [Brier, 1950], whose 1950 proper scoring rule for probabilistic weather forecasts is the intellectual origin of this line of work and is the reason that the statistical-forecast-evaluation vocabulary predates the finance literature by several decades.

Our per-target calibration buffer — adding the per-anchor empirical correction $b(\tau) \in \{0.045, 0.045, 0.020, 0.010\}$ at the four anchor targets and linearly interpolating off-grid before inverting the calibration surface — is heuristic. The principled replacement is conformal prediction. Vovk, Gammerman, and Shafer [Vovk et al., 2005] give the book-length foundation: under exchangeability, split-conformal prediction wraps any base predictor with a calibration-set correction that yields *finite-sample* marginal coverage at level τ. Shafer and Vovk [Shafer and Vovk, 2008] provide the standard tutorial exposition. Romano, Patterson, and Candès [Romano et al., 2019] combine conformal calibration with quantile regression, producing adaptive heteroscedastic intervals with the coverage guarantee preserved — this is the closest methodological match to the conformal upgrade we flag for future work, in which the log-log volatility regression becomes a quantile regressor and the calibration buffer becomes the conformal correction. Two recent developments are critical because our setting is neither exchangeable nor stationary: Barber, Candès, Ramdas, and Tibshirani [Barber et al., 2023] extend conformal prediction to non-exchangeable data using weighted quantiles and randomisation, establishing coverage guarantees under distribution drift; and Xu and Xie [Xu and Xie, 2021] develop EnbPI, a bootstrap-ensemble conformal interval for dynamic time-series with approximately valid marginal coverage in strongly-mixing settings. Either reference provides a concrete path from our empirical heuristic to a finite-sample guarantee, and we flag both as the upgrade route in §10.

2.4 Institutional model risk management

The final literature is not academic but regulatory, and it defines what "institutional-grade calibration disclosure" means. The foundational US supervisory guidance is SR Letter 11-7 [of Governors of the Federal Reserve System & Office of the Comptroller of the Currency, 2011], issued jointly by the Federal Reserve and the OCC in 2011, which articulates a framework for model development, implementation, validation, effective challenge, and ongoing monitoring for regulated financial institutions. Any bank integrating an on-chain oracle into a collateral-valuation or capital-computation pipeline is operating under this framework — and the framework explicitly requires validation evidence of the form we publish. Because our paper is being written in April 2026, we also cite SR Letter 26-2 [Board of Governors of the Federal Reserve System, 2026], the April 2026 revision that supersedes SR 11-7 for banking organisations over the $30B asset threshold and introduces a materiality-graduated validation construct. The revision updates expectations rather than replaces them; the validation and effective-challenge requirements we target are reaffirmed.

On the backtesting methodology specifically, the Basel Committee's 1996 *Supervisory framework for the use of "backtesting"* [on Banking Supervision, 1996] is the global floor. The framework defines green, yellow, and red zones for VaR backtesting based on exception counts over 250 observations, with regulatory responses escalating by zone. The mathematical content of the framework — an unconditional-coverage test over a rolling window — is the Kupiec [Kupiec, 1995] test we reproduce. That the same test appears in a 1995 academic paper, a 1996 supervisory framework, and a 2011 and 2026 pair of supervisory guidance letters is not an accident; it is what "institutional-grade calibration" has converged on, and it is the language we have adopted because protocol integrators and their regulators already speak it.

Section 2 length: ~2.9 pages in rendered markdown (approximately 1,700 words of body text across four subsections, with 40 citations distributed 12 / 12 / 13 / 3 across the four buckets — oracles / price-discovery+microstructure / calibration-conformal / model-risk — with the Basel backtest reference cross-cited in §2.3 and §2.4).

3 Problem Statement

3.1 Setup and notation

Let $\mathcal{S} = \{s_1, \ldots, s_K\}$ be a finite universe of assets (in our evaluation, ten US equities, ETFs, and hybrid crypto-proxies) and let $t \in \mathbb{N}$ index a sequence of *prediction windows* — in this paper, non-trading weekends defined by $[t_{\text{pub}}, t_{\text{tgt}}] =$ (Friday 16:00 ET, Monday 09:30 ET).

For each (s, t) we observe:

- $P_t(s) \in \mathbb{R}_{>0}$: the realised reference price at t_{tgt} (Monday open).
- $\mathcal{F}_t(s)$: pre-publish information available at t_{pub}. This includes $P_{t^-}(s)$ (Friday close), contemporaneous factor returns (E-mini S&P, gold, 10Y treasury, BTC futures), implied-vol indices (VIX, GVZ, MOVE), a calendar flag $\text{earn}_t(s) \in \{0, 1\}$ for an earnings release in the coming week, and a gap-length flag $\ell_t \in \{0, 1\}$ for calendar weekends ≥ 4 days.
- A *regime labeler* $\rho : \mathcal{F}_t(s) \to \mathcal{R}$ with $\mathcal{R} = \{\texttt{normal}, \texttt{long_weekend}, \texttt{high_vol}\}$, defined purely on pre-publish information.

A *base forecaster* f emits, at each (s, t, q) with claimed quantile $q \in (0, 1)$, a band $\big[L_t^f(s; q), U_t^f(s; q)\big]$.

3.2 The classical oracle problem

A conventional point-plus-CI oracle publishes, at each t_{pub}, a triple $(\hat{P}_t, L_t, U_t)$ with an *implicit* or *undisclosed* coverage level. The consumer has no contract on the realised rate at which $P_t \in [L_t, U_t]$. When CI widths are derived from publisher dispersion (Pyth) or a fixed Gaussian heuristic (Chainlink staleness fallback), the mapping from the CI to a realised probability-of-coverage statement is neither asserted nor auditable.

3.3 The coverage-inversion primitive

We define a stricter contract. Fix a base forecaster f, a regime labeler ρ, a discrete claimed-quantile grid $\mathcal{Q} \subset (0, 1)$, and a rolling historical dataset

$$\mathcal{D} = \big\{\big(P_t(s), \mathcal{F}_t(s), \{(L_t^f(s; q), U_t^f(s; q))\}_{q \in \mathcal{Q}}, \rho(\mathcal{F}_t(s))\big) : t \in \mathcal{T}_{\text{hist}}\big\}.$$

The **empirical calibration surface** is the mapping

$$S^f(s, r, q) \;=\; \frac{1}{|\mathcal{T}_{s,r}|} \sum_{t \in \mathcal{T}_{s,r}} \mathbf{1}\big[\, L_t^f(s; q) \leq P_t(s) \leq U_t^f(s; q) \,\big],$$

where $\mathcal{T}_{s,r} = \{t \in \mathcal{T}_{\text{hist}} : \text{symbol} = s,\ \rho(\mathcal{F}_t(s)) = r\}$.

Given a *consumer-chosen* target coverage $\tau \in (0, 1)$, the oracle serves the band at

$$q_{\text{served}}(s, r, \tau) \;=\; (S^f)^{-1}(s, r, \tau),$$

with $(S^f)^{-1}$ defined by linear interpolation between bracketing grid points and a documented fallback to a pooled surface $S^f(\cdot, r, q)$ when $|\mathcal{T}_{s,r}| < n_{\min}$.

The oracle read is the tuple

$$\texttt{PricePoint}(s, t, \tau) \;=\; \big(\hat{P}, L_t^f(s; q_{\text{served}}), U_t^f(s; q_{\text{served}}),\ q_{\text{served}},\ r,\ f\big),$$

exposing q_{served} and f as calibration *receipts* alongside the band itself.

3.4 Properties we claim

(P1) Auditability. $\mathcal{D}$ is reproducible from public data sources (freely available price feeds, vol indices, and earnings calendars). Given $(\mathcal{D}, f, \rho, \mathcal{Q})$, the surface S^f is deterministic; thus any third party can reconstruct S^f and independently verify that the served band corresponds to the published q_{served}.

(P2) Conditional empirical coverage. Under the assumption that the conditional distribution $P_t \mid (\mathcal{F}_t,\ \text{regime})$ is stationary across $\mathcal{T}_{\text{hist}}$ and the evaluation set, the realised coverage of the served band satisfies

$$\Pr\big[\, P_t \in [L_t^f(s; q_{\text{served}}),\ U_t^f(s; q_{\text{served}})] \;\big|\; \rho(\mathcal{F}_t) = r,\ s \,\big] \;\longrightarrow\; \tau$$

as $|\mathcal{T}_{s,r}| \to \infty$. Finite-sample deviations are measurable, tested (§6), and disclosed as a per-target *calibration buffer* applied pre-inversion (§4.3 specifies the deployed schedule).

(P3) Per-regime serving efficiency. For each regime r, the oracle selects $f^\star(r)$ from a candidate set $\mathcal{F}$. The deployed selection is the *empirically preferred deployment choice* on the evaluated sample under two criteria observable post-hoc on historical data: (i) mean bandwidth at matched realised coverage, and (ii) the Christoffersen independence p-value as a violation-clustering diagnostic. The Christoffersen p-value is a sample-size- and test-form-dependent diagnostic rather than a natural decision-theoretic optimisation target, so the property is an empirical-dominance statement on this sample under this evaluation battery, not a theorem-shaped optimality claim. In our evaluation $\mathcal{F} = \{\texttt{F0_stale},\ \texttt{F1_emp_regime}\}$; §7 reports the ablation matrix that justifies the hybrid regime-to-forecaster assignment. On held-out data at the deployed buffer the hybrid is *weakly dominant* over F1-everywhere on both criteria — bands tighten and the Christoffersen-independence margin widens, with no observed trade-off between the two — so the hybrid is the preferred deployment choice under the paper's evaluation objective. The in-sample motivation (mean-bandwidth dominance of F0 in `high_vol`) is preserved as the design heuristic; the out-of-sample evidence ratifies it on both criteria. This property is an oracle-serving claim about width allocation at matched realised coverage; it is not a claim that the resulting band is already the optimal liquidation-policy threshold for a lending protocol.

3.5 Non-goals

We explicitly do not claim:

1. **Optimal point estimates.** $\hat{P}$ is the midpoint of a band calibrated for coverage, not a minimum-variance estimator. We make no claim to improve over Pyth / Chainlink on point accuracy during market hours.
2. **Parametric tail guarantees.** The surface is empirical; shock-regime coverage is bounded above by the fraction of historical observations in the same tail of the forecast distribution (§9).
3. **Distribution-free coverage.** P2 requires stationarity of the conditional distribution. A conformalised variant (Vovk, Barber et al.) would upgrade the statement to a finite-sample guarantee under exchangeability; we flag this as future work.
4. **Protection against adversarial data feeds.** Soothsayer is a calibration-transparency primitive, not an integrity primitive. Its value proposition is orthogonal to oracle manipulation resistance; it is intended to be integrated *alongside*, not in place of, an adversarially-robust price feed.
5. **Optimal lending-policy defaults.** We do not claim in this paper to identify the welfare-optimal liquidation rule, collateral haircut, or target τ for a lending protocol. Those questions require explicit assumptions about account-weight distributions, protocol-specific loss functions, and the semantics by which realised outcomes are mapped into liquidation quality.

3.6 Evaluation questions

The body of the paper answers:

- **Q1** (calibration, §6). Does the served band achieve τ within 2pp on held-out data? What is the Kupiec unconditional-coverage p-value, and do violations cluster (Christoffersen independence p)?
- **Q2** (sharpness, §6–§7). How much tighter than a naïve stale-hold Gaussian baseline is the served band at matched realised coverage, per regime?
- **Q3** (ablations, §7). Which components of $f^\star$ (factor switchboard, log-log vol regression, earnings regressor, long-weekend regressor, hybrid regime selection, empirical buffer) are load-bearing for the P2/P3 claims, and which are disclosed for auditability but contribute no measurable signal at our sample size? The ablation reports block-bootstrap 95% CIs on each pairwise effect size.

- **Q4** (auditability, §4 + artifact). Can a third party reconstruct S^f on public data and independently verify a served `PricePoint`?

Questions such as "which served target τ should a Kamino-style protocol choose as its liquidation-policy default?" are downstream integration questions. They are important, but they sit outside Q1-Q4 and are not answered by this paper's evidence alone.

4 Methodology

This section instantiates the abstract setup of §3 with the concrete forecasters, surface, inversion procedure, and serving-time machinery used in our evaluation. The deployed implementation lives in `src/soothsayer/oracle.py` (Python) and `crates/soothsayer-oracle/src/{config,oracle}.rs` (Rust); the two are byte-for-byte verified by `scripts/verify_rust_oracle.py` (75/75 cases match) and we cite line-anchors in this section to make the artifact-paper correspondence explicit.

4.1 The forecaster pair $\mathcal{F}$

Our evaluation uses two base forecasters $\mathcal{F} = \{\texttt{F0_stale},\ \texttt{F1_emp_regime}\}$. Both emit, for every (s, t, q) with q on the claimed-coverage grid $\mathcal{Q}$ defined in §4.2, a band $[L_t^f(s; q),\ U_t^f(s; q)]$.

`F0_stale` — the stale-hold Gaussian baseline. Point estimate $\hat{P}_t^{\text{F0}} = P_{t^-}(s)$ (Friday close held forward). Conditional sigma $\sigma_t^{\text{F0}} = \hat{\sigma}_t^{\text{20d}} \cdot \sqrt{\ell_t} \cdot P_{t^-}(s)$ where $\hat{\sigma}_t^{\text{20d}}$ is the rolling 20-trading-day daily-return standard deviation at Friday close and $\sqrt{\ell_t}$ is a calendar-gap scaling ($\sqrt{1}$ for a standard Friday→Monday weekend, $\sqrt{2}$ for a 4-day weekend, $\sqrt{3}$ for 5-day, etc.). Bands are constructed parametrically: $L_t^{\text{F0}}(s; q) = \hat{P}_t^{\text{F0}} - z_q \cdot \sigma_t^{\text{F0}}$ and analogously for U, where $z_q = \Phi^{-1}\big(\frac{1+q}{2}\big)$. F0 is the methodological analog of a Chainlink-style stale `price` feed wrapped with a vol-scaled Gaussian — the "what does the simplest principled baseline look like" comparator.

`F1_emp_regime` — the factor-switchboard / log-log volatility-regime / empirical-quantile forecaster. Three components:

1. *Point estimate.* $\hat{P}_t^{\text{F1}} = P_{t^-}(s) \cdot (1 + r_t^{\text{factor}}(s))$ where $r_t^{\text{factor}}(s)$ is the weekend return of the per-symbol conditioning factor (E-mini S&P futures `ES=F` for the seven equities, gold futures `GC=F` for `GLD`, 10-year treasury futures `ZN=F` for `TLT`, BTC for `MSTR` post-2020-08). Section 5.4 specifies `FACTOR_BY_SYMBOL` in full.
2. *Conditional sigma.* For each (s, t) we fit a per-symbol log-log regression on the prior 156-weekend window:
$$\log|\varepsilon_\tau| \;=\; \alpha + \beta \log v_\tau(s) + \gamma_{\text{earn}} \cdot \text{earn}_\tau(s) + \gamma_{\text{long}} \cdot \ell_\tau \;+\; \xi_\tau, \quad \tau \in [t-156,\ t-1],$$
where $\varepsilon_\tau = \log\big(P_\tau(s)/\hat{P}_\tau^{\text{F1}}\big)$ is the return-residual of F1's point estimate and $v_\tau(s)$ is the per-symbol implied-volatility index (VIX for equities, GVZ for `GLD`, MOVE for `TLT` — see §5.4). Regressors with zero in-window variance are silently dropped from that fit (avoids singular matrices when a ticker has no in-window earnings observations). The implied conditional sigma at t is $\sigma_t^{\text{F1}} = \exp(\hat{\alpha} + \hat{\beta} \log v_t(s) + \hat{\gamma}_{\text{earn}} \cdot \text{earn}_t(s) + \hat{\gamma}_{\text{long}} \cdot \ell_t)$.
3. *Empirical-quantile bands.* In-window residuals are standardised by the regression-implied sigma, $z_\tau = \varepsilon_\tau / \sigma_\tau^{\text{F1, hist}}$. Bands at claimed quantile q are $L_t^{\text{F1}}(s; q) = \hat{P}_t^{\text{F1}} \cdot \exp\big(z_{(1-q)/2} \cdot \sigma_t^{\text{F1}}\big)$ and analogously for U at $z_{(1+q)/2}$, where $z_{(\cdot)}$ are empirical quantiles of the in-window standardised residuals. There is no Gaussian assumption on the residuals; non-zero median residuals propagate into the band without distortion (see §6.6 for the bias-absorption derivation). The earnings and long-weekend regressors enter as level shifts in $\log \sigma$ — they widen or tighten the conditional sigma deterministically without altering the residual distribution itself.

The pair $\mathcal{F}$ is deliberately small. We considered eight forecaster variants in early development (§7 reports the ablation matrix); the final pair survived empirical pruning. F0 is retained not as a deployed default but as the high-volatility fallback in the hybrid serving policy of §4.4.

4.2 Calibration surface and claimed-quantile grid

The claimed-coverage grid is $\mathcal{Q} = \{0.50, 0.60, 0.68, 0.75, 0.80, 0.85, 0.90, 0.925, 0.95, 0.975, 0.99, 0.995, 0.997, 0.999\}$ — fourteen points concentrated near the high-coverage end where consumers operate. The grid's upper tail was extended from a 0.995 cap to $\{0.997, 0.999\}$ in our 2026-04-25 revision after the buffer-tuning sweep showed serving at $\tau \geq 0.99$ benefited from finer grid resolution at the tail (see `reports/v1b_extended_grid.md`).

For each base forecaster $f \in \mathcal{F}$ we materialise the bounds at every $(s, t, q) \in \mathcal{S} \times \mathcal{T}_{\text{hist}} \times \mathcal{Q}$ into a long-form table persisted at `data/processed/v1b_bounds.parquet`. The empirical calibration surface

$$S^f(s, r, q) \;=\; \frac{1}{|\mathcal{T}_{s,r}|} \sum_{\tau \in \mathcal{T}_{s,r}} \mathbf{1}\big[\, L^f_\tau(s; q) \le P_\tau(s) \le U^f_\tau(s; q) \,\big]$$

is computed from this table, with $\mathcal{T}_{s,r} = \{\tau \in \mathcal{T}_{\text{hist}} : \text{symbol}(\tau) = s,\ \rho(\mathcal{F}_\tau(s)) = r\}$.

A pooled fallback surface $\bar{S}^f(\cdot, r, q)$, computed with the symbol axis collapsed, is materialised alongside. At serve time, the per-symbol surface is consulted first; the pooled surface is used when $|\mathcal{T}_{s,r}| < n_{\text{min}}$ for the active rolling window. The pooled-fallback path is exposed in the receipt as a `calibration` field with values `per_symbol` or `pooled`, so a consumer can identify which surface produced their served band.

4.3 Inversion, per-target buffer, and the served quantile

Given a consumer-chosen target $\tau \in (0, 1)$, the served claimed quantile is determined in three steps:

1. *Per-target buffer.* The consumer's τ is shifted by an empirical buffer:

   $$\tilde{\tau} \;=\; \min\big(\tau + b(\tau),\ \tau_{\text{max}}\big), \quad \tau_{\text{max}} = 0.999,$$

   where $b(\tau)$ is the value of the deployed buffer schedule:

   $$b \in \{(0.68 \mapsto 0.045),\ (0.85 \mapsto 0.045),\ (0.95 \mapsto 0.020),\ (0.99 \mapsto 0.010)\}$$

   with linear interpolation between anchors and flat extrapolation outside $[0.68, 0.99]$. The buffer is exposed in the receipt as `calibration_buffer_applied`. The four-anchor schedule was tuned on the OOS 2023+ slice as the smallest buffer per anchor satisfying realised-within-0.5pp of target plus Kupiec $p_{uc} > 0.10$ plus Christoffersen $p_{ind} > 0.05$ — see `reports/v1b_buffer_tune.md` for the sweep.
2. *Surface inversion.* The buffered target is mapped to a claimed quantile by inverting $S^f(s, r, \cdot)$. Off-grid targets are resolved by linear interpolation between the bracketing claimed-grid points. The bracketing pair is exposed in the diagnostic as `bracketed`.
3. *Bound lookup.* The served band is read from the bounds table at q_{served}. The two natural alternatives — interpolating the bounds rather than the surface, or rounding q_{served} to the nearest grid point and serving that band — were tested in early development; the surface-interpolation, bound-lookup approach was chosen because it preserves the deterministic correspondence between q_{served} (which is in the receipt) and the actual band edges (which the consumer can verify against the persisted bounds parquet).

4.4 Hybrid regime-to-forecaster policy

The serving layer consults a per-regime forecaster map:

`REGIME_FORECASTER` $=$ {`normal` $\to$ `F1_emp_regime`, `long_weekend` $\to$ `F1_emp_regime`, `high_vol` $\to$ `F0_stale`}.

The choice of $f^\star(r)$ for each r is justified empirically in §7 (in-sample bandwidth at matched coverage) and §6.4 (out-of-sample Christoffersen independence). The policy is static — it is fixed pre-deployment and exposed as a constant in `oracle.py:REGIME_FORECASTER` rather than being re-derived from data at serve time. A consumer reading the `regime` and `forecaster_used` fields of the `PricePoint` can verify the policy was applied as documented.

4.5 The PricePoint receipt

Every oracle read returns a `PricePoint` exposing nine fields beyond the band itself:

Field	Source	Used by
`target_coverage`	request	echoed for audit
`calibration_buffer_applied`	$b(\tau)$ from §4.3	reproducing the buffered $\tilde{\tau}$
`claimed_coverage_served`	q_{served} from §4.3	identifies the band's claimed quantile
`forecaster_used`	$f^\star(r)$ from §4.4	identifies which S^f was inverted
`regime`	$r = \rho(\mathcal{F}_t)$ from §5.5	identifies the surface bucket

Field	Source	Used by
`sharpness_bps`	$(U - L)/(2 \cdot P_{t^-})$ in bp	consumer's instantaneous capital-efficiency cost
`diagnostics.calibration`	`per_symbol` or `pooled`	identifies whether the fallback was used
`diagnostics.bracketed`	bracketing $\mathcal{Q}$ pair	verifies the inversion interpolation
`diagnostics.regime_forecaster_pollR`	`REGIME_FORECASTER`	snapshot of the deployed policy

Together these fields fully specify how a given (s, t, τ) request was resolved: a third party with the surface CSV (`reports/tables/v1b_calibration_surface.csv`) and the bounds parquet can reconstruct the served band from the receipt alone. This is the operational instantiation of property P1 (auditability) of §3.4.

4.6 Implementation and reproducibility

The deployed Oracle is implemented in two languages with byte-for-byte parity:

- *Python.* `src/soothsayer/oracle.py` — `Oracle.fair_value(symbol, as_of, target_coverage)`. Numeric parameters are exposed as module-level constants (`BUFFER_BY_TARGET`, `MAX_SERVED_TARGET`, `REGIME_FORECASTER`, `DEFAULT_FORECASTER`).
- *Rust.* `crates/soothsayer-oracle/src/{config,oracle}.rs` — same API, same constants, same arithmetic. The Rust port is the production serving binary; the Python is the research and backtest implementation. `scripts/verify_rust_oracle.py` runs both implementations on a 75-case test panel and asserts byte-for-byte equality on every field of the returned `PricePoint`. Parity was verified after every methodology change in the 2026-04-25 sequence (per-target buffer landing, grid extension, $\tau = 0.99$ buffer bump).

End-to-end reproduction is a single command per stage:

```
uv run python scripts/run_calibration.py    # rebuild bounds + surface artifacts
uv run python scripts/smoke_oracle.py        # cross-symbol demo of the served API
uv run python scripts/refresh_oos_validation.py  # produces the §6.4 OOS tables
```

All inputs are public and free (§5.7); no credentials are required for the calibration backtest itself.

5 Data and Regime Labeler

This section specifies the symbol universe, the weekend-panel construction, the per-symbol factor and volatility-index switchboards introduced in §4.1, the regime labeler ρ introduced in §3.1, and the train/test split underlying the §6 evaluation. All inputs are publicly available and free; the panel rebuilds end-to-end via `scripts/run_calibration.py`.

5.1 Symbol universe

We evaluate on $K = 10$ symbols spanning three asset classes:

Class	Symbols	Rationale
US equities	SPY, QQQ, AAPL, GOOGL, NVDA, TSLA, HOOD, MSTR	Eight underliers of the eight Solana xStocks for which Backed-issued tokenised equities trade on-chain
Tokenised gold	GLD	RWA-class generalisation: closed-market same as equities, distinct factor (gold futures) and vol index (GVZ)
Tokenised long rates	TLT	RWA-class generalisation: closed-market, distinct factor (10Y treasury futures) and vol index (MOVE)

The eight equity tickers are the underliers of the SPYx, QQQx, AAPLx, GOOGLx, NVDAx, TSLAx, HOODx, and MSTRx tokens deployed on Solana mainnet (Backed Finance, mid-2025 launch; mint addresses verified on-chain in `docs/v5-tape.md`). They are the immediate xStock-protocol consumer set. GLD and TLT are RWA generalisation anchors — they share the closed-market structure with the equities (NYSE hours) but exercise different factors and vol indices, so a coverage-inversion primitive that calibrates well across all three classes generalises to a broader closed-market RWA universe than equity-only evaluation would justify.

We do not use the on-chain xStock prices themselves as an input or evaluation target. Backtest target prices are the *underlying-venue* (NYSE) Monday opens, fetched from Yahoo Finance daily history. The on-chain xStock TWAP is a separate signal, deferred to v2 (`docs/v2.md` §V2.1) once the V5 tape (`docs/v5-tape.md`) accumulates 150 weekends of post-launch data.

5.2 Weekend panel construction

Each row of the panel is a single (s, t) weekend prediction window. For each symbol we walk the daily price history and, for every consecutive pair of trading days separated by a calendar gap of ≥ 3 days (`gap_days`), emit a row with:

- `fri_close` — close of the day before the gap (Friday or pre-holiday session)
- `mon_open` — open of the day after the gap (Monday or post-holiday session)
- `gap_days` — calendar days in the gap (3 for Friday→Monday, 4 for 3-day weekend, 5 for 4-day weekend, etc.)
- `fri_vol_20d` — rolling 20-trading-day standard deviation of daily log-returns at Friday close
- `factor_ret` — weekend return of the per-symbol conditioning factor (§5.4)
- `vol_idx_fri_close` — close of the per-symbol implied-volatility index at t_{pub} (§5.4)
- `earnings_next_week` — calendar flag from Yahoo Finance `earnings_dates` (1 if any earnings in the seven trading days following t_{pub}, else 0)
- `is_long_weekend` — 1 if `gap_days` ≥ 4, else 0

The panel spans 2014-01-17 (first Friday on which the rolling 20-day vol is defined for all symbols) through 2026-04-17 (last full weekend before paper compilation), yielding $|\mathcal{T}_{\text{hist}}| = 5,986$ rows across the 10 symbols. The MSTR series begins 2014-01-17; the HOOD series begins 2021-08-13 (post-IPO), contributing ~ 245 weekends.

5.3 Pre-publish features

All features used by either forecaster are observable at $t_pub = $ Friday 16:00 ET (or pre-holiday close). The full feature set:

- $P_{t^-}(s)$ — Friday close (or pre-holiday close, generalising)
- $r_t^{\text{factor}}(s)$ — weekend return $\log\big(F_t(s)/F_{t^-}(s)\big)$ of the per-symbol conditioning factor, where F is the futures or BTC contract for s
- $v_t(s)$ — close of the per-symbol implied-volatility index at t_{pub}
- $\hat{\sigma}_t^{\text{20d}}$ — rolling 20-trading-day standard deviation of daily log-returns at Friday close (used by F0; not used by F1)
- $\text{earn}_t(s) \in \{0, 1\}$ — earnings calendar flag for the post-publish week
- $\ell_t \in \{0, 1\}$ — long-weekend flag (`gap_days` ≥ 4)

The factor return $r_t^{\text{factor}}(s)$ requires the futures or BTC price at $t = $ Monday 09:30 ET. Globex-traded futures (`ES=F`, `NQ=F`, `GC=F`, `ZN=F`) and Bitcoin (`BTC-USD`) trade through the weekend, so $F_t(s)$ at Monday 09:30 is observable in real time at $t_{\text{pub}} + 65.5$h — well before Monday's NYSE open. In live deployment, the factor return is the *only* feature that requires post-publish computation; in the backtest it is read from cached daily history.

5.4 Factor and volatility-index switchboards

The per-symbol mappings are static and module-level constants in `src/soothsayer/backtest/panel.py`:

Symbol	Factor (`FACTOR_BY_SYMBOL`)	Vol index (`VOL_INDEX_BY_SYMBOL`)
SPY, QQQ, AAPL, GOOGL, NVDA, TSLA, HOOD	`ES=F` (E-mini S&P futures)	`^VIX`

Symbol	Factor (FACTOR_BY_SYMBOL)	Vol index (VOL_INDEX_BY_SYMBOL)
MSTR (pre 2020-08-01)	ES=F	^VIX
MSTR (2020-08-01 onward)	BTC-USD	^VIX
GLD	GC=F (gold futures)	^GVZ (CBOE Gold ETF Volatility Index)
TLT	ZN=F (10-year T-note futures)	^MOVE (ICE BofA MOVE Index)

The MSTR factor pivot at 2020-08-01 corresponds to MicroStrategy's first Bitcoin treasury purchase; pre-pivot MSTR's price-discovery linkage to broad equities dominated, post-pivot it pivots to BTC. The pivot date is hardcoded in panel.py:MSTR_BTC_PIVOT and applied row-wise during panel construction.

The vol-index choices are evidence-driven: an early V1b pass found that fitting the F1 log-log sigma regression with VIX as the volatility covariate yielded $\hat{\beta} \approx 0.55$ for GLD and $\hat{\beta} \approx 0.94$ for TLT — both substantially below the equity-class mean of $\hat{\beta} \approx 1.5$. This indicated VIX is a weak proxy for gold and treasury volatility. Substituting GVZ (gold-options-implied volatility) for GLD and MOVE (treasury-options-implied volatility) for TLT lifted $\hat{\beta}$ into the equity range and improved coverage at matched bandwidth. The asset-class-appropriate volatility index is one of two ways the methodology generalises beyond equities (the other being the factor switchboard).

5.5 The regime labeler ρ

The labeler $\rho : \mathcal{F}_t(s) \rightarrow \{\texttt{normal}, \texttt{long_weekend}, \texttt{high_vol}\}$ is implemented in src/soothsayer/backtest/regimes.py as a strict priority cascade:

1. high_vol — VIX at Friday close is in the top quartile of its trailing 252-trading-day window. Computed weekend-by-weekend from the VIX time series; the rolling-quartile threshold is itself observable at t_{pub} (uses VIX values strictly before the current Friday). The threshold is global (uses VIX even for non-equity symbols) — the regime is a market-condition flag, not a symbol-specific volatility measurement.
2. long_weekend — gap_days ≥ 4 (3-day weekend, 4-day weekend, holiday bridge). Applied only if high_vol did not already match.
3. normal — all other weekends.

Sample sizes per regime on the $5,986$-weekend panel:

Regime	n	Share
normal	3,924	65.6%
high_vol	1,432	23.9%
long_weekend	630	10.5%

A separate post-hoc tertile labeler tags each weekend by realised-move z-score (calm / normal / shock) using $|z| = |\log(P_t/P_{t^-})|/\hat{\sigma}_t^{20\text{d}}$. This realized_bucket is *not* a regime in the §3.1 sense — it depends on the realised target — and is used only for diagnostic stratification (the shock-tertile coverage ceiling reported in §9.2). It never enters ρ or the calibration surface.

We considered two refinements that were tested and dropped from the v1 deployment: a sub-regime split of normal into post_shock / calm / range_bound, and an FOMC/CPI/NFP macro-event regressor (reports/v1b_macro_regressor.md). Neither lifted shock-tertile coverage measurably; the implied-vol indices already absorb the macro-uncertainty signal those refinements were designed to capture. Sub-regime granularity is retained as a v2 candidate; the empirical motivation for prioritising it is in §10.

5.6 Train/test split

The split date is **2023-01-01**: weekends with fri_ts < 2023-01-01 are the calibration set, the remainder are the held-out OOS evaluation set.

Split	Rows	Weekends	Ticker-coverage
Calibration (2014-01 → 2022-12)	4,266	466	All 10 (HOOD partial; begins 2021-08)
Held-out OOS (2023-01 → 2026-04)	1,720	172	All 10

The split is temporally disjoint at the surface-construction level. The calibration surface S^f is built from the 4,266 calibration-set rows (per-(symbol, regime, forecaster, claimed-quantile) cells) and held *fixed* throughout the OOS evaluation. Every OOS row is served through the frozen surface: no parameter, threshold, or grid point is updated using OOS data. The per-target buffer schedule (§4.3) was tuned on this same OOS slice using the methodology of `reports/v1b_buffer_tune.md` — this is disclosed as a (then) sample-size-one calibration step in §9.4 and partially closed by the walk-forward stability evidence (`reports/v1b_walkforward.md`, summarised in §9.4) showing the deployed buffer values land at the cross-split mean.

The 2023-01-01 split point is conservative in two respects: (i) it places the 2023 banking turbulence and the 2024–2025 macro-regime transition in the held-out slice, exposing the surface to material out-of-sample regime shifts; (ii) it leaves only ~ 245 HOOD rows in the calibration set (HOOD IPO'd in 2021), so the per-symbol surface for HOOD is the leanest of all ten symbols and exercises the pooled-fallback path more often than the others — a stress test for the fallback design.

5.7 Provenance and reproducibility

All inputs are pulled live from Yahoo Finance via the `yfinance` Python library (`src/soothsayer/sources/yahoo.py`); pulls are cached to local parquet under `data/raw/yahoo_*.parquet` to insulate the backtest from API rate limits and intermittent provider unavailability. No credentials are required for any Phase 0 data source. The `earnings_dates` flag uses the same library's earnings-calendar endpoint.

The end-to-end calibration backtest runs under fifteen minutes on a 2024 M3 MacBook from a cold cache, and under one minute when the daily-history cache is warm. Reproduction:

```
uv sync
cp .env.example .env       # optional; only required for live on-chain workloads
uv run python scripts/run_calibration.py
```

The single script materialises `data/processed/v1b_bounds.parquet` (the bounds table at every (s, t, q)), `reports/tables/v1b_calibration_surface.csv` and `v1b_calibration_surface_pooled.csv` (the per-symbol and pooled surfaces), and refreshes the §6 OOS-evaluation tables. A consumer in possession of these three artifacts can independently verify any served `PricePoint` against the receipt fields specified in §4.5.

6 Results

This section reports the primary calibration evidence. Sharpness and per-component effects are deferred to §7 (ablations); system-level implementation and audit reproducibility are in §8.

6.1 Evaluation protocol

The backtest panel comprises $N = 5,986$ weekend prediction windows: ten symbols (SPY, QQQ, AAPL, GOOGL, NVDA, TSLA, HOOD, GLD, TLT, MSTR) observed across 638 weekends from 2014-01-17 through 2026-04-17. Each window runs Friday 16:00 ET → Monday 09:30 ET. Monday open is the target $P_t(s)$; Friday close and contemporaneous factor-futures / vol-index readings compose the pre-publish information $\mathcal{F}_t(s)$.

For the evaluation we report two complementary figures:

- **In-sample machinery check.** Calibration surface S^f fit on the full panel; Oracle served on the same panel. Realised coverage should be close to target τ *by construction* — the inversion is against the surface that was fit on this data. A material deviation indicates a bug in the surface-build or inversion code path, not a real-world failure mode.
- **Out-of-sample validation (the real number).** Calibration surface fit on $\mathcal{T}_{\text{train}} = \{t : t < \text{2023-01-01}\}$ (4,266 weekend rows); Oracle served on $\mathcal{T}_{\text{test}} = \{t : t \geq \text{2023-01-01}\}$ (1,720 rows, 172 weekends). No information

from $\mathcal{T}_{\text{test}}$ enters the surface. This is the closest analog to a production deployment in which the surface is rebuilt on a rolling schedule.

All reported p-values are for two-sided Kupiec unconditional-coverage tests (POF likelihood-ratio, χ^2_1 null) and Christoffersen independence tests (χ^2 with k degrees of freedom where k is the number of per-symbol time series with both violation and non-violation observations).

6.2 Raw forecaster coverage (motivation for the calibration surface)

Before serving-time inversion, the base forecasters are not calibrated at their nominal claimed-coverage levels. At claim $q = 0.95$ on the full panel:

Forecaster	Realised	Half-width (bps)	MAE (bps)
F0 (stale-hold + Gaussian)	0.976	401.8	94.8
F1_emp_regime (factor + log-log + regressors)	0.923	253.7	90.0

F0 over-covers (97.6% vs 95% claimed) at the cost of blanket-width bands, as is expected from an uninformative Gaussian with no factor adjustment. F1_emp_regime under-covers at the nominal 95% claim by 2.7pp. This is not a deployment path — it is the input to the calibration surface. Neither raw forecaster is the product. The product is $\mathcal{S}^f$ and the served band at q_{served}.

6.3 Served-band calibration — in-sample machinery check

Consumer target $\tau \in \{0.68, 0.95, 0.99\}$; full surface; full panel ($N = 5,986$):

Regime (n)	$\tau = 0.68$: realised / width (bps)	$\tau = 0.95$: realised / width	$\tau = 0.99$: realised / width
normal (3,924)	0.725 / 97.2	0.978 / 341.5	0.980 / 384.9
long_weekend (630)	0.719 / 117.6	0.970 / 389.2	0.981 / 429.4
high_vol (1,432)	0.735 / 174.6	0.974 / 585.6	0.985 / 700.1
pooled (5,986)	**0.727 / 117.8**	**0.976 / 404.9**	**0.981 / 465.0**

Pooled realised coverage at $\tau = 0.95$ is 0.976 — a $+2.6$pp over-coverage against the consumer request. The in-sample Kupiec test rejects at every τ; this is expected because the inversion uses a discrete 12-point claimed grid and a small positive buffer on the target, both of which bias the served band toward over-coverage. *In-sample over-coverage is the safe direction;* under-coverage would indicate an implementation defect.

6.4 Served-band calibration — out-of-sample (2023+)

Calibration surface fit on pre-2023 bounds; Oracle served on 2023+ weekends ($N_{\text{test}} = 1,720$ rows, 172 weekends). The calibration surface is held-out — frozen on the 4,266 pre-2023 calibration rows; no parameter, threshold, or grid point is updated using OOS data. The per-anchor empirical buffer schedule of §4.3 is *deployment-tuned* on this same OOS slice via the methodology of `reports/v1b_buffer_tune.md`. The numbers below are therefore best read as a deployment-calibrated operating result on a held-out surface, not as a clean held-out validation end-to-end. The walk-forward stability evidence in the inset block below is the empirical mitigation of the buffer-OOS-tuning concern.

Regime (n)	$\tau = 0.68$: realised / width	$\tau = 0.85$: realised / width	$\tau = 0.95$: realised / width	$\tau = 0.99$: realised / width
normal (1,150)	0.691 / 121.2	0.863 / 229.7	0.945 / 417.7	0.977 / 502.8
long_weekend (190)	0.632 / 121.0	0.821 / 283.3	0.953 / 416.6	0.958 / 465.2
high_vol (380)	0.663 / 188.0	0.850 / 299.6	0.963 / 591.6	0.984 / 874.5
pooled (1,720)	**0.678 / 135.9**	**0.855 / 251.1**	**0.950 / 456.0**	**0.977 / 580.8**

6.4.1 Conditional-coverage tests (pooled OOS)

τ	Violations	Rate	Kupiec LR	p_{uc}	Christoffersen LR	p_{ind}
0.680	553	0.322	0.018	**0.893**	7.818	**0.647**
0.850	249	0.145	0.373	**0.541**	13.750	**0.185**
0.950	**86**	**0.050**	**0.000**	**1.000**	9.500	**0.485**
0.990	40	0.023	22.224	0.000	3.801	0.956

The $\tau = 0.95$ row is the primary oracle-validation operating result on the deployed configuration. On the held-out surface served at the deployment-tuned per-anchor buffer, the Oracle delivers realised coverage of exactly 0.950 — Kupiec $p_{uc} = 1.000$ (test statistic essentially zero, no evidence of mis-calibration) and Christoffersen $p_{ind} = 0.485$ (no clustering of violations). The conjunction passes by margin, not by inches. The number is *deployment-calibrated* on the OOS slice rather than purely held-out end-to-end, and should be read in conjunction with the walk-forward stability evidence below and §9.4 (the wider disclosure of the buffer-tuning provenance).

Walk-forward stability of the deployed buffer. Because the per-anchor buffer schedule of §4.3 is fit on the same OOS slice that this section evaluates, "deployment-calibrated" rather than "purely held-out" is the strongest defensible characterisation of the table above. The relevant empirical mitigation is the cross-split distribution of the buffer itself. We re-run the buffer-tuning sweep on six expanding-window train/test splits spanning 2019-01 through 2025-01 (`reports/v1b_walkforward.md`), each split fitting the surface and the per-anchor buffer on its own train side and reporting the buffer that satisfies the same realised-within-0.5pp + Kupiec $p_{uc} > 0.10$ + Christoffersen $p_{ind} > 0.05$ criterion that selected the deployed schedule. At $\tau = 0.95$, the cross-split mean is 0.019 ($\sigma = 0.017$); the deployed value 0.020 lands at the cross-split mean to within rounding. At $\tau = 0.85$, the cross-split mean is 0.025 ($\sigma = 0.022$) and the deployed value 0.045 is conservative ($\geq 1\sigma$ above the mean). The deployed buffers are therefore not idiosyncratic to the 2023+ slice in the sense that a multi-split rerun reproduces the same operating point at $\tau = 0.95$ and a tighter operating point at $\tau = 0.85$. This evidence is what allows the §6.4 number to be characterised as "deployment-calibrated and walk-forward-stable" rather than "potentially overfit to one slice"; it does not upgrade the result to a purely held-out end-to-end validation, which remains §10.1's V2.2 (rolling calibration-surface rebuild).

This result should be read as validation of the oracle's coverage contract at $\tau = 0.95$, not as proof that $\tau = 0.95$ is the welfare-optimal operating point for a protocol that consumes the band for liquidations or collateral haircuts.

Other operating points. At $\tau = 0.85$ (the current protocol-deployment default; downstream welfare comparisons against protocol baselines are the subject of Paper 3, which revisits the earlier simplified Kamino-style flat-band benchmark against the production xStocks reserve configuration and oracle semantics), realised coverage is 0.855, both tests pass at $\alpha = 0.05$. At $\tau = 0.68$, realised lands 0.2pp below target with both tests passing comfortably. At $\tau = 0.99$, Kupiec rejects: realised coverage is 0.977, materially below the requested 0.99. This is a structural ceiling — the rolling 156-weekend per-(symbol, regime) calibration window cannot resolve the 1% tail reliably; even with the bounds grid extended to 0.999 (`reports/v1b_extended_grid.md`), the per-bucket sample size is too small to estimate the 1% quantile with the precision the consumer is asking for. The Christoffersen independence test continues to pass on the OOS slice ($p_{\text{ind}} = 0.956$): the rejected coverage is a level-attribution failure, not a violation-clustering failure. §9.1 discusses the structural ceiling; §10 frames the conformal-prediction upgrade that would address it.

The Oracle therefore passes Kupiec + Christoffersen at three of four standard operating points (0.68, 0.85, 0.95) on held-out data, with the upper-tail failure (0.99) disclosed as a finite-sample structural ceiling rather than a deployment defect.

6.4.2 Extended diagnostics — Berkowitz, DQ, CRPS, exceedance magnitude

The Kupiec and Christoffersen tests of §6.4 are the regulatory standard but they are not the entirety of the modern density-forecast diagnostic kit. We additionally report four reviewer-likely tests on the same OOS panel; full numerical artefacts in `reports/tables/v1b_oos_reviewer_diagnostics.csv` and `reports/tables/v1b_oos_berkowitz_crps.csv`.

Berkowitz (2001) joint LR on inverse-normal-transformed PITs. Continuous PITs are reconstructed per row by interpolating the served-band CDF, evaluated at a 19-point central-quantile grid ($\{0.05, 0.10, \ldots, 0.95\}$). The Berkowitz test rejects (LR $= 36.10$, $p \approx 0$). The rejection is driven by a variance contraction in the inverse-normal-transformed PITs ($\widehat{\text{var}}\, z = 0.84$ vs the unit variance under H, mean $\hat{\mu} = 0.07$, AR(1) $\hat{\rho} = -0.04$). Mechanism: the per-target buffer schedule of §4 is calibrated for the four anchor $\{0.68, 0.85, 0.95, 0.99\}$ and flat-extrapolates outside that range. At low

(e.g. = 0.05) the flat-extrapolated buffer (0.045) is over-conservative by design, producing systematic over-coverage in the central region of the predicted distribution and the variance contraction Berkowitz detects. The test is therefore correctly identifying the deployment choice: the served band is per-anchor calibrated, not full-distribution calibrated, and the deviation outside the deployment range [0.68, 0.99] is in the *safe direction* (over-coverage, not under-coverage). The reliability diagram in `reports/figures/v1b_reliability_diagram.png` (left panel) makes this concrete: the four anchor land on the diagonal; the inter-anchor central- grid points lie above the diagonal in the < 0.55 region and on the diagonal above = 0.65.

Engle-Manganelli (2004) Dynamic Quantile (DQ) test. Per-symbol DQ regressions of the hit indicator on four lags + intercept, summed into a pooled $\chi^2(50)$ statistic across the ten symbols (mirroring the Christoffersen pooling of §6.4). Results: $p_{\text{DQ}} = 0.033$ at = 0.68, $p_{\text{DQ}} = 0.014$ at = 0.85, $p_{\text{DQ}} = 0.032$ at = 0.95, $p_{\text{DQ}} \approx 0$ at = 0.99. DQ rejects at all four anchors at = 0.05 — moderately at 0.95 (statistic just above the $\chi^2(50)$ critical value of 67.5) and strongly at = 0.99. This is a stricter test than Christoffersen's two-state Markov-chain independence test of §6.4 and detects multi-lag conditional structure that the simpler test misses. Caveat: the per-symbol-pooled DQ inflates with the number of symbols; running per-symbol and reporting the median p-value is an alternative aggregation that we report in the appendix as a sensitivity check. The DQ rejection at 0.95 is a disclosure rather than a deployment block — Christoffersen p_{ind} remains comfortably above the 0.05 threshold (§6.4) and the realised rate matches the target by Kupiec.

Continuous Ranked Probability Score (CRPS). Mean CRPS across the OOS panel is 1.82 in price units (median 0.97). This is reported as a baseline number for cross-oracle comparators in future work (§10); the absolute value is not interpretable without a competing forecaster on the same panel. CRPS decomposes into reliability + resolution + uncertainty per Gneiting-Raftery (2007); the per-regime CRPS decomposition is in `reports/tables/v1b_oos_pit_continuous.csv`.

McNeil-Frey-style exceedance magnitude. When the served band misses, *by how much*. Distribution of breach sizes (basis points of Friday close) per :

	Violations	Mean (bps)	Median (bps)	p_{95} (bps)	Max (bps)
0.68	553	104	49	416	2,339
0.85	249	120	50	456	2,150
0.95	86	125	56	465	1,415
0.99	40	131	72	469	796

The maximum breach *shrinks monotonically* in (2,339 → 2,150 → 1,415 → 796 bps) — wider bands catch larger shocks, as one would hope. Critically for the = 0.99 structural-ceiling disclosure of §9.1: the 40 missed violations have median breach 72 bps and max 796 bps. The level-attribution failure at = 0.99 (realised 0.977 vs target 0.99) is therefore not a tail-blowup story; the misses are bounded by ~800 bps and median 72 bps. A protocol consuming the = 0.99 band on a 100% LTV position would experience a worst-case 8% mismatch on a missed event, with median ~0.7%. This is a quantitative bound on the practical cost of the structural ceiling and is, in our reading, materially weaker than the qualitative "structural ceiling" framing of §9.1 alone suggests.

The four extended diagnostics together support a refined paper claim: the served-band Oracle is *per-anchor calibrated* (Kupiec + Christoffersen at the four anchors, §6.4), not *full-distribution calibrated* (Berkowitz rejects, this section). The deviation from full-distribution calibration is in the safe direction (over-coverage outside the deployment range), and the worst-case practical impact at the = 0.99 structural ceiling is bounded by 800 bps maximum breach.

6.5 Per-symbol generalisation

Per-symbol realised coverage of the served band at $\tau = 0.95$ on the full in-sample panel (the OOS subsample of 172 weekends is too small per symbol for ticker-level reads):

Symbol	Realised	Half-width (bps)
AAPL	0.929	257.9
GLD	0.928	142.2
GOOGL	0.932	193.4
HOOD	0.877	391.3
MSTR	0.901	390.4

Symbol	Realised	Half-width (bps)
NVDA	0.923	319.1
QQQ	0.930	172.2
SPY	0.935	193.4
TLT	0.917	142.2
TSLA	0.930	319.1

These are *raw-forecaster* realised coverages at the nominal 0.95 claim — not served-band. They confirm the pattern motivating the calibration-surface approach: the raw forecaster under-covers at the nominal level, and the amount of under-coverage varies meaningfully across tickers (HOOD: 87.7%, MSTR: 90.1%, SPY: 93.5%). The calibration surface stratified by (symbol, regime) absorbs this variation. The aggregate OOS pass in §6.4 confirms that the surface + inversion mechanism reconciles this heterogeneity.

6.6 Point accuracy and bias absorption

Point-estimate MAE on the full panel: 90.0 bps for F1_emp_regime vs 94.8 bps for F0 stale-hold. RMSE: 163.3 bps vs 176.4 bps. Median bias on the *raw* factor-adjusted point: -5.2 bps; on the *raw* F0 stale point: -4.1 bps. With residual standard deviation $\sigma \approx 163$ bps and $n = 5,986$, the factor-adjusted bias has $t \approx -2.5$, which is statistically distinguishable from zero. The factor-adjusted point delivers a 5% MAE improvement.

A natural reading of the bias figure is that the served Oracle inherits a calibration defect from its raw point. It does not, and the construction is worth making explicit. The empirical-quantile architecture takes quantiles of the *log-residual* distribution

$$\varepsilon_t = \log(P_t/\hat{P}_t)$$

directly — no zero-mean assumption is made. The lower and upper bounds are constructed as $\hat{P}_t \cdot \exp(z_{\text{lo}})$ and $\hat{P}_t \cdot \exp(z_{\text{hi}})$, where $z_{\text{lo}}, z_{\text{hi}}$ are the empirical quantiles of ε_t on the rolling window. If ε_t has a non-zero median, that asymmetry shifts z_{lo} and z_{hi} by the same amount, and the served band is bias-aware by construction. The served point used at the API surface — defined as $(L_t + U_t)/2$ — is therefore the midpoint of a bias-corrected band, not the raw factor-adjusted estimate $\hat{P}_t$. The -5.2 bps figure is a property of the *raw forecaster*, made transparent here for completeness; it does not propagate to the consumer-facing point or to the coverage figures of §6.4.

A dedicated point-forecasting benchmark against incumbent oracles, on a matched intraday + closed-market window, is deferred to future work (§10).

6.7 Summary

On the deployed configuration evaluated on the 2023+ held-out slice at consumer target $\tau = 0.95$:

- Realised coverage: 95.0%
- Kupiec $p_{uc} = 1.000$ (pass)
- Christoffersen $p_{ind} = 0.485$ (pass)
- Mean served half-width: 456 bps; per-regime widths: 417.7 bps (normal), 416.6 bps (long_weekend), 591.6 bps (high_vol)
- Walk-forward buffer stability (§6.4): six-split cross-split mean at $\tau = 0.95$ is 0.019 ($\sigma = 0.017$); deployed value 0.020 lands at the mean.

The number is best read as *deployment-calibrated and walk-forward-stable* rather than purely held-out end-to-end: the calibration surface is held-out, the per-anchor buffer is deployment-tuned on the same slice, and a six-split walk-forward ratifies the deployed buffer at $\tau = 0.95$ as the cross-split mean.

Extended diagnostics (§6.4.1): Berkowitz rejects (driven by buffer-induced over-coverage outside the deployment range — safe direction), DQ rejects at all four anchors (multi-lag conditional structure beyond Christoffersen's two-state independence test, disclosed), CRPS = 1.82 (baseline for future cross-oracle comparators), exceedance magnitude bounded by 800 bps max at = 0.99 with median 72 bps.

These figures constitute the empirical support for claim P2 (conditional empirical coverage, §3.4). Claim P3 (per-regime serving efficiency) is supported by the ablation reported in §7. Claim P1 (auditability) is evidenced by the artifact

released with this paper (§8). None of the figures in this section, by themselves, identify an optimal liquidation-policy default for a lending protocol; they validate the oracle contract that an integrator would then have to map into its own loss function.

7 Ablation Study

This section answers Q3 of §3.6: which components of `F1_emp_regime` and the serving layer are load-bearing for the calibration claim, and which are disclosed for auditability but contribute no measurable signal at our sample size? We report two ablation studies — an eight-step *forecaster ladder* on the full panel (§7.1–§7.3) and a five-cell *serving-layer matrix* on the held-out 2023+ slice (§7.4) — and conclude with a load-bearing-vs-cosmetic taxonomy (§7.5). All deltas carry block-bootstrap 95% confidence intervals; the bootstrap unit is the weekend (`fri_ts`), with all symbol-rows sharing a weekend resampled together so cross-sectional correlation is preserved (1000 resamples). Raw tables: `reports/tables/v1b_ablation_*.csv`.

7.1 Ladder design

Each ladder rung adds exactly one knob to the previous rung; the ablation isolates the marginal contribution of each design choice. Variants are scored on the full 5,986-weekend panel (2014-01-17 → 2026-04-17) at the nominal claimed quantile $q = 0.95$ — the *raw-forecaster* coverage, prior to the calibration-surface inversion of §4.2–§4.3.

Variant	Description	Realised	Mean half-width (bps)	Point MAE (bps)
A0	F0_stale: Friday close + 20-day Gaussian band	0.976	401.8	94.8
A1	+ factor-adjusted point + empirical residual quantile	0.914	244.4	90.0
A2	+ VIX-scaled residual standardisation	0.921	239.8	90.0
A3	+ log-log VIX regression on $\log \lVert\varepsilon\rVert$	0.923	252.6	90.0
A4	+ per-symbol vol index (VIX / GVZ / MOVE)	0.924	253.5	90.0
A5	+ earnings-next-week regressor (only)	0.924	253.8	90.0
A6	+ is_long_weekend regressor (only)	0.924	253.5	90.0
A7	full `F1_emp_regime` (both regressors)	0.923	253.7	90.0
B0	A7's CI width centred on stale point	0.919	253.7	94.8

Every variant's *raw-forecaster* Kupiec test rejects at the nominal 0.95 claim — this is by design. The 0.95 figure here is the un-buffered, un-inverted quantile-grid output; the served-band coverage of §6.4 is produced by the calibration-surface inversion + per-target buffer that wraps any of these variants. The ablation isolates *which underlying components* drive the sharpness and coverage of the inversion's input, not the inversion's output.

7.2 Pooled effect — which knob delivers the sharpness?

A → B	Δ coverage (pp)	Δ sharpness (%)
A0 → A1 (factor pt + emp Q)	**6.2 [7.4, 5.0]**	**39.3 [41.3, 37.3]**
A1 → A2 (VIX-scale residuals)	+0.7 [0.1, +1.4]	1.9 [4.6, +1.0]
A2 → A3 (log-log VIX regression)	+0.3 [0.3, +0.9]	**+3.9 [+1.6, +6.4]**
A3 → A4 (per-symbol vol index)	+0.1 [0.1, +0.4]	**+0.3 [+0.1, +0.5]**
A4 → A5 (earnings regressor)	0.0 [0.1, +0.1]	+0.1 [0.2, +0.5]
A4 → A6 (long-weekend flag)	0.0 [0.3, +0.3]	0.0 [0.4, +0.5]
A4 → A7 (both regressors)	0.1 [0.4, +0.2]	+0.1 [0.5, +0.7]
A7 → B0 (strip factor pt; hold CI fixed)	0.4 [1.2, +0.4]	0.0 (by construction)

Negative Δ sharpness means tighter bands. Bold rows are those whose 95% CI excludes zero.

The factor-switchboard point + empirical residual quantile (A0 → A1) deliver essentially all of the pooled sharpness gain: bands are 39.3% tighter (CI [37.3%, 41.3%]) at the cost of 6.2pp lower raw-forecaster coverage. Every subsequent rung's pooled sharpness effect is either flat or a small widening. The earnings regressor (A4 → A5) and the long-weekend flag (A4 → A6) have no statistically detectable pooled effect — both confidence intervals span zero on coverage and sharpness.

The B0 variant strips the factor-adjusted point while holding A7's empirical CI width *constant* and centring on the stale point estimate. Pooled Δ coverage is 0.4pp (CI [1.2, +0.4]) — not statistically distinguishable from A7. The MAE rises 4.8 bps (94.8 vs 90.0). The factor-switchboard point's contribution is therefore concentrated in *point accuracy*, not in served-band coverage; at $n = 5,986$ the coverage contribution of the point shift is below detection.

7.3 Per-regime effect — where the regime-specific knobs earn their keep

The pooled view obscures three regime-specific findings.

A → B	regime	Δ coverage (pp)	Δ sharpness (%)
A0 → A1 (factor pt + emp Q)	normal	**3.0 [3.9, 2.2]**	**29.9 [32.3, 27.5]**
	long_weekend	**6.0 [8.6, 3.9]**	**50.7 [55.0, 46.1]**
	high_vol	**14.5 [18.3, 11.0]**	**52.1 [55.3, 49.0]**
A1 → A2 (VIX-scale)	normal	**1.5 [2.2, 0.8]**	**15.1 [17.1, 13.3]**
	long_weekend	1.9 [3.6, 0.0]	**16.6 [21.6, 11.2]**
	high_vol	**+7.2 [+5.3, +9.6]**	**+41.1 [+34.2, +48.7]**
A2 → A3 (log-log)	normal	**0.7 [1.2, 0.2]**	**2.2 [3.1, 1.2]**
	long_weekend	0.5 [2.3, +1.1]	1.7 [4.3, +0.8]
	high_vol	**+3.2 [+1.4, +5.1]**	**+15.6 [+9.6, +22.0]**
A4 → A6 (long-weekend flag)	normal	0.1 [0.4, +0.1]	**1.8 [2.1, 1.5]**
	long_weekend	+1.1 [0.3, +2.4]	**+10.6 [+8.5, +12.9]**
	high_vol	0.1 [0.6, +0.4]	+0.1 [0.6, +0.9]

Three observations.

1. **The log-log VIX regression (A1 → A3 cumulatively) exists for the `high_vol` regime.** It lifts high-vol coverage by 10.4pp (cumulative across A2 + A3) at the cost of 65% wider bands — the insurance trade a consumer wants in elevated-volatility weekends. In `normal` regime the same transition *costs* coverage by 2.2pp at no sharpness gain, which is why the §4.4 hybrid policy assigns `high_vol` to F0_stale rather than to A7.
2. **The long-weekend flag is a localised effect.** It widens bands by +10.6% **only in its own regime** (`long_weekend`), with statistically-flat effects in `normal` and `high_vol`. The flag costs nothing outside its regime and earns its keep inside it.
3. **The earnings regressor is not detectable in any regime.** Every cell of A4 → A5 has a 95% CI containing zero on both coverage and sharpness. A weekend-ahead binary flag is too coarse to carry signal through the empirical-quantile band-construction at $n = 5,986$. We retain it for *auditability* — every PricePoint receipt

records whether the flag was active — but not for performance, and we flag it as a v2 candidate for replacement with a finer event-granularity dataset (scheduled datetime + implied-move).

7.4 Serving-layer matrix — hybrid policy and calibration buffer

The forecaster ladder of §7.1–§7.3 measures the *raw-forecaster* contribution of each knob. The serving layer adds two additional design choices: the per-regime forecaster selection (§4.4) and the per-target empirical buffer (§4.3). We isolate these by re-serving the same OOS 2023+ panel (1,720 rows × 172 weekends) at the §6 headline target $\tau = 0.95$ across five (forecaster policy × buffer) cells under the deployed configuration. The calibration surface is fit on the pre-2023 calibration set (4,266 rows, 466 weekends) and held fixed; the deployed buffer at $\tau = 0.95$ is 0.020 per `BUFFER_BY_TARGET` (§4.3).

Cell	Forecaster policy	Buffer	Realised	Half-width (bps)	Kupiec p_{uc}	Christoffersen p_{ind}
C0	F1 everywhere	0.000	0.922	359.7	0.000	0.145
C1	F0 everywhere	0.000	0.921	293.2	0.000	0.474
C2	hybrid (F1 normal/LW, F0 high_vol)	0.000	0.920	351.0	0.000	0.380
C3	F1 everywhere	0.020	0.949	467.9	0.826	0.221
C4	**hybrid (deployed Oracle)**	**0.020**	**0.950**	**456.0**	**1.000**	**0.485**

Comparison	Δ coverage (pp)	Δ sharpness (%)	Interpretation
C0 → C2 (hybrid effect, no buffer)	0.1 [0.7, +0.4]	2.4 [5.6, +1.0]	Hybrid alone doesn't change realised coverage; saves ~2% bandwidth
C0 → C3 (buffer effect, no hybrid)	**+2.7 [+1.9, +3.7]**	**+30.0 [+28.8, +31.4]**	Buffer closes OOS coverage gap at ~30% wider bands
C2 → C4 (buffer effect, with hybrid)	**+3.0 [+2.0, +3.9]**	**+29.9 [+28.6, +31.3]**	Buffer effect is approximately independent of hybrid
C0 → C4 (total serving layer)	**+2.9 [+2.0, +3.8]**	**+26.8 [+22.3, +31.3]**	Full Oracle delivers target coverage; pays ~27% sharpness vs raw F1

Negative Δ sharpness means tighter bands; CIs are 95% block-bootstrap by weekend over 1,000 resamples (`reports/tables/v1b_serving_ablation_bootstrap.csv`).

Three findings.

1. **The per-target buffer is the load-bearing serving-layer knob for coverage.** $+2.7$ to $+3.0$pp of the OOS gap is closed by the buffer at $\tau = 0.95$; the hybrid regime policy alone closes essentially none of it. Without the buffer, surface inversion delivers 0.92 realised against a 0.95 target and Kupiec rejects at every cell.
2. **The hybrid regime policy is the empirically preferred deployment choice on the evaluated sample, not a binary necessity.** At the deployed buffer, both F1-everywhere (C3) and the hybrid (C4) pass Kupiec and Christoffersen at $\alpha = 0.05$. The hybrid is *weakly dominant* over F1-everywhere on both evaluation axes: bands tighten from $467.9 \rightarrow 456.0$ bps (-2.6%), Kupiec margin tightens from $p_{uc} = 0.826$ to $p_{uc} = 1.000$,

and Christoffersen margin tightens from $p_{ind} = 0.221$ to $p_{ind} = 0.485$, with no observed trade-off between the two criteria. This is the empirical content of property P3 of §3.4 — *per-regime serving efficiency* defined as empirical dominance on the evaluated sample under (i) mean bandwidth at matched realised coverage and (ii) the Christoffersen independence p-value as a violation-clustering diagnostic (not as a decision-theoretic optimisation target). The hybrid is the preferred deployment choice under the paper's evaluation objective.

3. **The buffer effect is approximately additive to the hybrid effect.** The buffer-vs-no-buffer delta is $+2.7$pp without hybrid (C0 $\rightarrow$ C3) and $+3.0$pp with hybrid (C2 $\rightarrow$ C4) — bracketing CIs overlap, so we cannot reject buffer-hybrid additivity at $n = 172$ OOS weekends. Practically: deciding hybrid-vs-no-hybrid does not affect the buffer-tuning decision, and vice versa, which simplifies the schedule re-fitting required by any V2.2 rolling rebuild (§10.1).

The historical record of the v1 development sequence — including an earlier scalar 0.025 buffer that produced cleanly-different numbers, in particular a Christoffersen rejection at the F1-everywhere cell — is preserved in `reports/v1b_ablation.md` and the methodology evolution log at `reports/methodology_history.md`.

7.5 Load-bearing-vs-cosmetic taxonomy

Synthesising §7.2–§7.4:

Component	Role	Evidence
Factor-switchboard point + empirical residual quantile (A0 $\rightarrow$ A1)	**Load-bearing for sharpness.** Pooled 39.3% tighter at $n = 5,986$.	§7.2
VIX-scaled residual standardisation + log-log VIX regression (A1 $\rightarrow$ A3)	**Load-bearing for `high_vol` coverage.** Cumulative +10.4pp coverage in `high_vol` at the cost of 65% wider bands — the insurance trade.	§7.3
Per-symbol vol index (VIX / GVZ / MOVE)	**Load-bearing for non-equity RWA generalisation.** The 0.3% pooled bandwidth effect is statistically resolved but small; the substantive justification is the GLD/TLT $\hat{\beta}$ recovery documented in §5.4.	§5.4, §7.2
Long-weekend flag	**Load-bearing inside `long_weekend` regime, neutral outside.** Targeted +10.6% widening only when the regime fires.	§7.3
Earnings-next-week flag	**Disclosed for auditability; no detectable performance contribution at our sample size.** Candidate for v2 replacement with finer event granularity.	§7.3
Empirical calibration buffer (per-target schedule)	**Load-bearing for OOS coverage.** +2.9pp coverage closes the OOS gap at the deployed buffer; without it, surface inversion alone delivers raw realised 0.92 at $\tau = 0.95$ and Kupiec rejects.	§7.4
Hybrid regime-to-forecaster policy	**Empirically preferred deployment choice on the evaluated sample; weakly dominant over F1-everywhere on (sharpness, Christoffersen margin).** At deployed buffer: tighter bands (-2.6% vs F1-everywhere) and larger Christoffersen p_{ind} margin (0.485 vs 0.221), with no observed trade-off.	§7.4

Two components above (the per-symbol vol index and the earnings flag) sit at the boundary of statistical resolution and are disclosed accordingly. The remainder are load-bearing, with each contribution attributed to a specific regime or serving-layer property and bracketed by a bootstrap CI that excludes zero. The reviewer-facing claim of §3.4 — that the deployed methodology is fit for purpose, with each component justified by either bootstrap evidence (load-bearing) or transparent disclosure (cosmetic) — is supported component-by-component by this ablation.

8 Serving-Layer System

The methodology of §4 is implemented as a four-component serving stack designed for two requirements that pull in opposite directions: (i) the model logic is research-iterated in Python and re-validated against the §6 OOS panel after every change; (ii) the production hot path is a Solana program reading bands at consumer-facing latency. This section describes how those requirements are reconciled in a single source of truth, and how the serving-time output of the Rust + on-chain stack is contractually identical to the Python reference's.

8.1 Split-by-function architecture

We separate *training* (Python) from *serving* (Rust + Solana).

```
Python (model + train)                Rust (serve)
----------------------               ------------------
scripts/run_calibration.py             soothsayer-oracle (lib)
  -> v1b_bounds.parquet      -->       -> Oracle::fair_value()
  -> v1b_calibration_surface.csv          (hybrid + buffer + inversion)
  -> v1b_calibration_surface_pooled.csv
                                       soothsayer-publisher (CLI)
                                       -> prepare-publish -> on-chain payload

                                       soothsayer-oracle-program (Anchor)
                                       -> publish() -> PriceUpdate PDA

                                       soothsayer-consumer (no_std SDK)
                                       -> decode_price_update(&[u8])
```

The Python `Oracle` (`src/soothsayer/oracle.py`) is the canonical specification. It both (a) produces the calibration artefacts consumed by every downstream serving component and (b) is the executable reference for the inversion algorithm. A change to the methodology — a new buffer schedule, a new regime, a grid extension — is implemented in Python first, validated end-to-end against the §6 OOS panel, and then ported to Rust under the byte-for-byte parity contract of §8.5. No model logic is duplicated between the two implementations: the training pipeline materialises the surface and bounds table once offline; the serving stack reads them.

8.2 The Rust crate stack

Four crates plus one Anchor program, with deliberately narrow responsibilities.

Crate	Purpose	Lines	Tests
`soothsayer-oracle`	Library: loads the Python artefacts, exposes `Oracle::fair_value` with byte-exact equivalence to Python	~600	included in parity harness
`soothsayer-publisher`	CLI binary `soothsayer`: serves single reads, lists available `(symbol, fri_ts)` pairs, prepares on-chain publish payloads	~500	6/6 unit tests

Crate	Purpose	Lines	Tests
`soothsayer-consumer`	`no_std`, minimal-dep SDK for downstream protocols: decodes the on-chain `PriceUpdate` PDA into a typed `PriceBand` view	~300	5/5 unit tests
`soothsayer-demo-kamino`	Reference Kamino-style consumer integration: consumes a `PriceBand` and emits `Safe / Caution / Liquidate` decisions	~400	included in integration tests
`soothsayer-oracle-program` (Anchor)	On-chain program: `initialize`, `publish`, `set_paused`, `rotate_signer_set`	~400	scaffolded, devnet deployment one command away

The publisher CLI's primary subcommand is `fair-value`; its secondary subcommand `prepare-publish` converts an off-chain `PricePoint` (Python's nine-field receipt of §4.5) into the on-chain `PublishPayload` wire format of §8.4. The consumer SDK is intentionally `no_std` so it can be vendored into a Solana program account-decoder without dragging in the Rust standard library.

8.3 On-chain program

`programs/soothsayer-oracle-program/` is a vanilla Anchor program with three PDA account types and four instructions.

Accounts.

- `PublisherConfig` — global PDA seeded `[b"config"]`. Carries the authority pubkey, an emergency-pause flag, and the minimum-publish-interval rate-limit.
- `SignerSet` — global PDA seeded `[b"signer_set"]`. For Phase 1 the "Merkle root" is just the single publisher's pubkey stored directly; the multi-replicator Merkle-verification path is scaffolded for Phase 3.
- `PriceUpdate` — per-symbol PDA seeded by the symbol bytes. One account per symbol in the universe; first publish creates the account (`init_if_needed`), subsequent publishes overwrite in place.

Instructions.

- `initialize` — one-time setup; creates `PublisherConfig` and `SignerSet` PDAs.
- `publish(payload: PublishPayload)` — validates the signer membership, enforces the cadence rate-limit, checks on-chain band-ordering and coverage-bound invariants, and writes the new `PriceUpdate`. Emits a structured Anchor `Published` event.
- `set_paused` — authority-only emergency stop.
- `rotate_signer_set` — authority-only signer rotation.

Every instruction emits an Anchor event so off-chain indexers can trace publish history without re-reading every account state. The `publish` instruction enforces five on-chain invariants before persisting: band ordering ($L \leq \hat{P} \leq U$), coverage values $\leq 10,000$ basis points, exponent $\in [-12, 0]$, signer membership in the `SignerSet`, and time since the last publish $\geq$ `min_publish_interval_secs`.

8.4 Wire-format design

The `PriceUpdate` account layout is informed by the Pyth-Network September-2021 BTC flash-crash post-mortem (publicly disclosed; see external references). Three design rules avoid the failure modes that incident exposed.

```
PriceUpdate (128 bytes data + 8-byte Anchor discriminator):

  version              u8         schema version, currently 1
  regime_code          u8         0=normal, 1=long_weekend, 2=high_vol
  forecaster_code      u8         0=F1_emp_regime, 1=F0_stale
  exponent             i8         shared across price fields, default -8
  _pad0                [u8; 4]

  target_coverage_bps  u16        consumer's request, integer bp
  claimed_served_bps   u16        served quantile, integer bp
  buffer_applied_bps   u16        empirical buffer, integer bp
  _pad1                [u8; 2]

  symbol               [u8; 16]   ASCII null-padded

  point                i64        absolute fixed-point at `exponent`
  lower                i64        absolute fixed-point at `exponent`
  upper                i64        absolute fixed-point at `exponent`
  fri_close            i64        absolute fixed-point at `exponent`

  fri_ts               i64        unix timestamp of anchor Friday 20:00 UTC
  publish_ts           i64        unix timestamp of this publish
  publish_slot         u64

  signer               Pubkey     32 bytes
  signer_epoch         u64
```

The three rules:

1. **Absolute prices, not deltas.** A consumer reading `lower` gets a real price directly. There is no `point + lower_delta` arithmetic that a downstream integration could get wrong.
2. **Single shared exponent.** The same `exponent` field applies to `point`, `lower`, `upper`, and `fri_close`, so cross-field comparisons are exact integer comparisons after a single multiply. There is no per-field-exponent mismatch footgun.
3. **Integer basis points, never floats.** Coverage and buffer values are `u16` basis points (9,500 = 95.00%). This is exact and deterministic across Rust, the Anchor IDL, TypeScript clients, and downstream Python consumers — no float-parse or float-compare drift between layers.

The full publish payload serialises to 66 bytes; the on-chain account is 128 bytes after Anchor padding. Both sizes are locked by unit tests in `crates/soothsayer-publisher/src/payload.rs` so a future schema change cannot silently break the wire format.

8.5 Byte-for-byte parity verification

`scripts/verify_rust_oracle.py` runs the Python `Oracle.fair_value` and the Rust `soothsayer fair-value` binary on the same `(symbol, fri_ts, target_coverage)` triples and asserts byte-exact agreement on the numeric output (point, lower, upper, sharpness_bps, claimed_served, calibration_buffer_applied) plus exact agreement on the string fields (regime, forecaster_used, calibration, clipped, bracketed). The current panel covers 75 cases: a deterministic sample of 25 random `(symbol, fri_ts)` pairs from the bounds table, plus explicit edge cases for SPY, GLD, TLT, MSTR, and HOOD at their most-recent Friday, served at three targets each ($\tau \in \{0.68, 0.95, 0.99\}$). **75/75 cases pass** as of the most recent re-run.

The parity harness is invoked after every methodology change. The 2026-04-25 sequence — per-target buffer landing (replacing the scalar 0.025), grid extension to {0.997, 0.999} with `MAX_SERVED_TARGET = 0.999`, and the $\tau = 0.99$ buffer bump from 0.005 to 0.010 — was applied to Python first, then ported to Rust; in each case the parity harness was re-run and re-verified before the change was considered shipped. The harness is the operational guarantee that the on-chain published bands correspond to the methodology evaluated in §6 and ablated in §7. It is also the contract that allows a consumer to verify a served `PricePoint` against the published surface artefact without trusting either the Python or the Rust implementation in isolation: the two implementations produce identical output on identical inputs.

8.6 Reproduction and end-to-end serve path

End-to-end reproduction is a small number of commands per stage:

```
# Train (Python, ~1 min warm cache, ~15 min cold)
uv sync
uv run python scripts/run_calibration.py

# Serve (Rust, single read)
cargo build --release -p soothsayer-publisher
./target/release/soothsayer fair-value --symbol SPY --as-of 2026-04-17 --target 0.85

# Verify Rust <-> Python parity (~30 s)
PYTHONUNBUFFERED=1 uv run python scripts/verify_rust_oracle.py

# Deploy to devnet (after a one-time Solana + Anchor toolchain install)
./scripts/deploy_devnet.sh
```

The on-chain publish step is a separate concern handled by `soothsayer prepare-publish` (which produces the borsh-encoded `PublishPayload` bytes) plus a TypeScript signer that calls the program's `publish` instruction. The TS integration harness lives at `tests/integration.ts` and is exercised against `solana-test-validator` for local development; the devnet path is exercised by `scripts/deploy_devnet.sh`. A lending-protocol integrator wanting to consume the on-chain band depends on `soothsayer-consumer` (the `no_std` decoder) and reads the `PriceUpdate` PDA at the symbol's derived address; the demo Kamino-style consumer at `crates/soothsayer-demo-kamino/` shows the full integration in ~400 lines, including the LTV decision logic that uses the band's lower bound (the conservative reading) rather than the point estimate as the collateral valuation input.

The design of the serving stack is therefore a contract between three audiences: (a) the *researcher* iterating methodology against the Python OOS panel; (b) the *protocol integrator* building against the on-chain `PriceUpdate` accounts; and (c) the *third-party auditor* verifying that a served band corresponds to the published surface. The byte-for-byte parity contract of §8.5 makes (a) and (b) identical at the numeric level, while the receipt fields of §4.5 plus the public artefacts of §4.6 make (c) executable from public data alone.

9 Limitations

This section enumerates the assumptions under which the claims of §3.4 hold, the failure modes the backtest exposed, and the domain boundaries outside which we make no assertion. Most items here correspond to explicit deferrals rather than unknown unknowns; each points forward to §10 (Future Work) with a concrete follow-up. In particular, the strongest open issues in the present draft — heuristic buffer tuning, no live deployment window, and the stricter Berkowitz / DQ rejections of §6.4.1 — are not contradictions of the coverage-inversion primitive. They are the empirical map of what a V2 system should improve.

9.1 The high-τ tail ceiling

At consumer target $\tau = 0.99$, the served Oracle delivers realised coverage 0.977 — a 1.3pp shortfall by Kupiec ($p_{uc} \approx 0$, LR $= 22.2$), but with quantitatively bounded impact. The exceedance-magnitude diagnostic of §6.4.1 reports that the 40 missed-coverage events at $\tau = 0.99$ have median breach 72 bps (of Friday's close) and maximum 796 bps; across the four anchor τ the maximum breach shrinks monotonically in τ ($2,339 \to 2,150 \to 1,415 \to 796$ bps for $\tau = 0.68 \to 0.85 \to 0.95 \to 0.99$), as wider bands catch larger shocks — the qualitative property a coverage-claiming oracle should have. The shortfall is therefore better characterised as a *level-attribution failure with bounded magnitude* than as an open-ended tail risk: the served band rate is wrong by 1.3pp at the highest anchor, and Kupiec correctly identifies this, but a protocol consuming the $\tau = 0.99$ band experiences a worst-case $\sim 8\%$ price-mismatch on missed events, with median $\sim 0.7\%$.

The structural cause is the per-(symbol, regime) calibration-window size, not the grid spacing. Our log-log residual model is fit on a rolling 156-weekend window; reliably resolving the 1% tail of that distribution requires more observations than any one bucket carries. The pooled-surface fallback partially mitigates this but does not eliminate it: the pooled bucket draws on ~ 600 observations per regime, still tight for 1-in-100 events. The 2026-04-25 extension of the bounds grid from a 0.995 cap to $\{0.997, 0.999\}$ lifted realised at $\tau = 0.99$ from 0.972 to 0.977 but did not close the gap, confirming sample-size as the binding constraint (see `reports/v1b_extended_grid.md`).

This is a finite-sample artefact of the empirical quantile estimator at the tail of the calibration window, not a defect of the coverage-inversion primitive. A conformalised upgrade (see §9.4) would report finite-sample coverage bounds instead, at the cost of assumption strength; protocols that require true $\tau = 0.99$ behaviour today should layer an exogenous tail-widening rule on top of the served band, calibrated to their own loss function.

9.2 Shock-tertile empirical ceiling

The weekend-realised-move distribution has a heavy tail: a realised-move z-score tertile labelled `shock` in our backtest (post-hoc, using $|z| \geq 0.67$ standardised by Friday's 20-day vol) has a structural realised-coverage ceiling of approximately 80% at the nominal 95% claim — because pre-publish information is insufficient to predict those moves. We do not attempt to raise this ceiling; we disclose it. Protocols that accept Soothsayer as a collateral-valuation input should treat shock-tertile tails as explicitly out-of-model and either apply their own exogenous widening (e.g., a circuit breaker tied to overnight futures moves) or reduce collateralisation during periods where shock-tertile entry is plausible.

The ceiling is reported transparently in every `PricePoint` diagnostic via the calibration receipt: the served q carries the historical realised coverage at its bucket, and a consumer can read off the tail risk before using the band.

9.3 Stationarity assumption behind P2

Claim P2 (conditional empirical coverage, §3.4) is an *empirical* statement. It holds under the assumption that the conditional distribution $P_t \mid (\mathcal{F}_t,\ \rho(\mathcal{F}_t) = r)$ is approximately stationary across $\mathcal{T}_{\text{hist}}$ and the target deployment horizon. This assumption is plausibly violated in at least three ways:

- **Structural breaks in microstructure.** The 2020 COVID shock, the 2022 rate regime, and the introduction of 24/7 tokenized-stock trading on Solana in mid-2025 each plausibly shifted the conditional distribution. Our 2023+ OOS slice includes the 2022-→-2023 regime transition; it does not cover a comparable future break.
- **Regime-labeler drift.** ρ uses a trailing 252-trading-day VIX percentile to tag `high_vol`; a persistent rise in baseline vol (e.g., 2022-like) shifts the threshold endogenously. This is a feature for rolling deployment but can defeat the cross-regime fairness of the `high_vol`-vs-`normal` comparison.
- **Factor-switchboard drift.** MSTR's factor mapping pivots from ES=F to BTC-USD on 2020-08-01 (the BTC-proxy transition). This is a hand-coded pivot in `FACTOR_BY_SYMBOL`. A second pivot — e.g., if MSTR's BTC sensitivity decays — would require a re-fit.

A conformalised variant (§9.4 and §10) would upgrade P2's guarantee from asymptotic-under-stationarity to finite-sample-under-exchangeability; the latter is strictly weaker and measurable.

9.4 Heuristic per-target calibration buffer

The served Oracle applies an empirical buffer to the consumer target before surface inversion, persisted in `BUFFER_BY_TARGET` and looked up by linear interpolation off-grid. The deployed values are $\{\tau{=}0.68 \to 0.045,\ \tau{=}0.85 \to 0.045,\ \tau = 0.95 \to 0.020,\ \tau = 0.99 \to 0.010\}$, each chosen as the smallest buffer satisfying realised coverage within 0.5pp of target with both Kupiec $p_{uc} > 0.10$ and Christoffersen $p_{ind} > 0.05$ on the OOS 2023+ slice (`reports/v1b_buffer_tune.md`); the $\tau = 0.99$ value was bumped from 0.005 to 0.010 in the 2026-04-25 grid extension to {0.997, 0.999}. The buffer is *load-bearing* for the OOS Kupiec pass at every $\tau \leq 0.95$; without it, the surface inversion delivers 92.2% realised at target 0.95 and Kupiec rejects.

Three weaknesses of this choice:

1. **Sample-size-one buffer (partially closed by walk-forward).** Each per-target buffer was originally fit on one train/test split. A six-split expanding-window walk-forward (2019–2025; `reports/v1b_walkforward.md`) produces a distribution of buffers across splits: at $\tau = 0.95$ the cross-split mean is 0.019 ($\sigma = 0.017$) and the deployed value 0.020 lands at the mean; at $\tau = 0.85$ the cross-split mean is 0.025 ($\sigma = 0.022$) and the deployed 0.045 is conservative ($\geq 1\sigma$ above mean). The disclosure is therefore tightened, not eliminated: we commit to re-measuring the schedule as part of any rolling calibration-surface rebuild (§10.1, V2.2).
2. **Per-target schedule, not a continuous function.** Off-grid targets interpolate linearly between the four anchor points. A reviewer might prefer a continuous function fit (e.g., spline) on a finer sweep grid; we have not done that comparison.
3. **No finite-sample guarantee.** The buffer is a heuristic, not a theorem. We tested vanilla split-conformal (operationally equivalent at our calibration size), Barber et al. nexCP-style recency-weighted conformal at two half-lives, and a block-recency baseline at two windows; all three under-corrected the OOS gap

relative to the per-target heuristic, with bootstrap 95% CIs that exclude zero. The full evidence is in `reports/v1b_conformal_comparison.md`. The conformal upgrade therefore is *not* an obvious win on this data; it is reserved as a v2 direction subject to either a finer claimed-coverage grid (above 0.995, removing the structural ceiling) or a multi-split walk-forward evaluation that distinguishes drift from sampling noise.

The choice to ship a per-target heuristic now and re-evaluate conformal in v2 is disclosed, not evaded. The `calibration_buffer_applied` field on every `PricePoint` is the first-class receipt — the consumer sees exactly which buffer was applied to their served band.

This matters for interpretation. The heuristic-buffer limitation is not that the current system is unusable; it is that the current correction is empirical rather than theorem-backed. The practical research question for V2 is therefore not "whether a buffer is needed" — §7 already shows that it is load-bearing — but which replacement improves on the current heuristic under live drift: a continuous correction fit, a conformal wrapper, or a rolling-reestimated schedule.

9.5 The earnings regressor is a disclosure, not a contribution

The `F1_emp_regime` base forecaster includes a 0/1 `earnings_next_week` flag among its log-log regressors. The ablation in §7 shows that this regressor has no detectable effect on coverage or sharpness at our sample size (Δcov $=$ 0.0pp, CI $[-0.1, +0.1]$; Δsharp% $= +0.1\%$, CI $[-0.2, +0.5]$). The regressor was included in anticipation of earnings-week volatility premia; at $N = 5,986$ and a weekly-granularity flag it is not detectable.

We retain it for two reasons: (a) it is part of the audit receipt — removing it would silently change the surface, and we consider surface stability across documented methodology a product property; (b) a future iteration may replace the weekly flag with a finer event-granularity dataset (earnings date + estimated move), in which case the structural slot is already in place. A paper reviewer who prefers the minimal model should feel free to read the ablation as authorising its removal.

9.6 Hybrid regime policy is empirically preferred on this sample, not a binary necessity

In-sample evidence (matched-realised-coverage sharpness dominance of F0 in `high_vol`) motivated the hybrid serving policy. Out-of-sample at the deployed buffer, the hybrid's mean-coverage effect is statistically indistinguishable from F1-everywhere (Δcov $= -0.1$pp, CI $[-0.7, +0.4]$). At the deployed buffer both F1-everywhere and the hybrid pass Kupiec and Christoffersen at $\alpha = 0.05$; the hybrid is *not* a binary verdict-flip on Christoffersen independence — it is the *empirically preferred deployment choice* on the evaluated sample, *weakly dominating* F1-everywhere on (mean bandwidth, Christoffersen margin) on both axes simultaneously with no observed trade-off: bands tighten by 2.6% ($467.9 \rightarrow 456.0$ bps) and the Christoffersen p_{ind} margin lifts from 0.221 to 0.485. The Christoffersen p-value is a violation-clustering diagnostic that is sample-size- and test-form-dependent rather than a natural decision-theoretic optimisation target, so this is an empirical-dominance statement under the paper's evaluation battery, not a theorem-shaped optimality claim. We disclose this softening of the original "Christoffersen independence is load-bearing" framing because at an earlier stage of v1 development (under a scalar 0.025 buffer) F1 + buffer did fail Christoffersen independence and the hybrid was strictly necessary; at the deployed buffer the hybrid is a margin-and-sharpness improvement rather than a defect-fix. The historical record of the buffer-tuning sequence is preserved in `reports/methodology_history.md`.

9.7 No live deployment window

The empirical claims of §6 are backtested, not live. As of the submission of this paper, the shipped Rust serving layer and on-chain publisher program (§8) are functionally verified (byte-for-byte PythonRust parity on 75 test cases) but have not run as a live price feed consumed by a production protocol. A live window — with real integrators, real consumers, and an adversarial environment — will exhibit modes that a backtest cannot. The paper should be read as *validation of a calibration primitive on historical data*, not as a production-quality assurance.

This limitation is orthogonal to the stricter diagnostic rejections above. A live deployment window would close the external-validity gap and let us measure drift, consumer-experienced coverage, and rebuild stability in production. It would not by itself make Berkowitz or DQ pass; those failures speak to the current statistical design of the calibration layer and regime structure, which is why §10 separates live-deployment work from methodology upgrades.

9.8 No numerical incumbent benchmark

We do not report a systematic numerical comparison of Soothsayer's served bands against Pyth, Chainlink Data Streams, or RedStone Live on a matched evaluation window. The conceptual comparison in §1.1 (stale-hold,

dispersion CI, undisclosed methodology) is qualitative; a numerical benchmark would require either access to incumbent-internal historical bands or a reconstruction of incumbent values from on-chain feeds. Our team has partial Chainlink v10 / v11 reconstruction infrastructure in-repo that was used for a sanity-check comparison (`reports/v1_chainlink_bias.md`, `reports/v1b_chainlink_comparison.md`) and a Pyth Hermes-based comparison (`reports/v1b_pyth_comparison.md`) but neither is at the rigor required for a research-paper-grade benchmark. The RedStone-side equivalent has additional limits documented in §9.8.1; a forward-cursor tape against the public RedStone gateway began continuous capture on 2026-04-26 (`scripts/run_redstone_scrape.py`, persisted to `data/processed/redstone_live_tape.parquet`) and will be reportable once it accumulates a paper-grade weekend sample. We flag the matched-window comparator dashboard as future work (§10) — specifically, an AFT-community-facing public comparator computed on a common historical window.

9.8.1 RedStone-specific scoping limits

Three structural facts further constrain what a RedStone Live comparison can claim, beyond the general incumbent-benchmark limitation above. (i) RedStone's public REST gateway (`api.redstone.finance/prices`) prices the *underlier tickers* (SPY, QQQ, MSTR), not the actual tokenized SPL xStock tokens (SPYx, QQQx, MSTRx) that consumers on Solana hold and that this paper's calibration surface targets — a comparison reads RedStone's view of SPY against our view of SPYx, which are economically related but not the same instrument. (ii) Coverage on the gateway is partial as of 2026-04-25: TSLA, NVDA, HOOD, and GOOGL return empty and AAPL was 33d stale, restricting any matched comparison to the SPY/QQQ/MSTR subset. (iii) Gateway retention is hard-capped at 30 days, which means a parity comparison against our 12-year underlying-equity panel is impossible — only forward-collected weekends from 2026-04-26 onward can enter such a comparison. Whether the paid RedStone Live WebSocket tier serves a confidence field, a longer history, or the missing equity tickers is not verifiable from public artifacts. The "undisclosed methodology" framing of §1 and §2 is therefore restated more precisely as "no calibration claim exposed in any public artifact (REST gateway, on-chain PDA, or launch post)."

9.9 No optimal liquidation-policy benchmark

The paper validates an oracle interface, not a lending policy. We do not report a decision-theoretic benchmark of "what target τ should a protocol choose?" or "does a regime-demoted liquidation threshold dominate a flat threshold once portfolio weights and protocol losses are specified?" Answering those questions requires at least three ingredients absent from the present evaluation: (i) an explicit distribution over borrower-book LTV weights rather than a pure coverage metric, (ii) a protocol-specific cost model distinguishing unnecessary liquidation, unnecessary caution, and missed liquidation / bad debt, and (iii) a declared semantics for what counts as the correct action under realised prices. Without those ingredients, a claim of optimal liquidation-policy default would be stronger than the evidence we present here.

9.10 Scope of the methodology — what fits and what does not

Region scope. All ten tickers in the backtest are US-listed. The weekend we predict is the US weekend. The factor-switchboard mapping (ES/NQ/GC/ZN/BTC) privileges US session data. We make no claim about the generalisation of the coverage-inversion primitive to tokenized-JP equities, tokenized-EU equities, or commodities whose primary discovery venue is not a US exchange. A multi-region replication is future work (§10.5).

Asset-class scope. The methodology requires a *continuous off-hours information set* — a public, free, sub-daily price signal that is observably correlated with the closed-market underlier. Tokenized US equities have this (E-mini index futures + VIX), tokenized gold has it (gold futures + GVZ), tokenized US treasuries have it (10Y T-note futures + MOVE), and most major FX pairs have it (overseas-session prices). Tokenized credit has it through CDS spreads and index volatility, modulo data licensing. We claim the methodology generalises to any RWA class with such a signal.

What the methodology does *not* fit, and we do not claim:

- **Real estate.** No continuous off-hours information set exists; NAV updates are discrete and infrequent. The closed-market problem here is structurally different (no signal, not "stale signal") and requires a different methodology.
- **Illiquid commodities.** Some agricultural and specialty commodities have futures markets but those markets are discontinuous or thinly traded. The factor regression's $\hat{\beta}$ would be unstable.
- **Instruments with primarily NAV-based pricing.** Money-market-style tokens whose price is administratively set rather than market-observed have no "fair value" question to calibrate against.

`docs/methodology_scope.md` in the artifact released with this paper carries the per-class fit/no-fit table that protocol integrators and risk teams can use as a quick filter. The broader claim of §1 — that calibration-transparent risk reporting is the missing infrastructure layer for responsible institutional-scale RWA onboarding — applies to the substantial subset of RWA classes within scope, and explicitly does *not* claim coverage for classes outside it.

9.11 On-chain tokenized-stock prices as an unused signal

Cong et al. [Cong et al., 2025] document that off-hour returns on tokenized stocks *anticipate*, rather than amplify, subsequent Monday opens. Their finding implies that the contemporaneous on-chain xStock price during a closed-market interval already aggregates a non-trivial part of the weekend information set. Our base forecaster does not read this signal: $\hat{P}$ is a function of off-chain factor returns and a per-symbol volatility index only. A natural competing baseline — and one a reviewer is likely to raise — is "use the on-chain xStock TWAP as the point estimate."

We do not report this comparison in the present paper for one reason: data history. Tokenized-equity primary venues on Solana launched in mid-2025, providing on the order of 30 weekends of on-chain price data through the cutoff of this evaluation. A stable per-(symbol, regime) Kupiec test requires on the order of 150 observations; per-regime Christoffersen requires more. The xStock-anchored forecaster is documented as a v2 deliverable in `docs/v2.md` §V2.1 (the *F_tok* forecaster, gated on the V5 on-chain tape that began continuous capture on 2026-04-24) and will be reported in the v2 paper once data history supports OOS validation. The omission is a deliberate evaluation-power choice, not an architectural exclusion.

9.12 MEV and consumer-experienced coverage

The realised coverage figures reported in §6 are computed against the realised Monday opening reference price P_t on the underlying venue. Daian et al. [Daian et al., 2020] document that on-chain transaction ordering is adversarial: searchers and validators compete to reorder transactions for profit, and band-edge events are economically attractive targets. A consumer who reads the Soothsayer band at on-chain mid and then transacts at a worse price (because of front-running, sandwich attacks, or block-builder reordering) experiences an *effective* coverage rate that may differ from our reported coverage rate near the band edge.

We do not currently measure this gap. The infrastructure to do so requires (i) a Solana tape with mempool-or-bundle granularity (the V5 forward-cursor tape, started 2026-04-24, supplies this but with insufficient history at submission), (ii) a model of consumer transaction behaviour at band-edge thresholds, and (iii) a Jito-bundle reconstruction of bundled flow. The MEV-adjusted coverage measurement is documented as v2 work in `docs/v2.md` §V2.3. The present paper's coverage claim should therefore be read at face value — coverage of P_t at the venue — not as a guarantee on consumer-experienced execution near the band edge.

9.13 What is not a limitation

For completeness, we enumerate two concerns that do *not* apply to the coverage-inversion primitive as specified:

- **Oracle manipulation.** Coverage transparency is orthogonal to integrity. Soothsayer is designed to be read alongside an adversarially-robust feed, not as its replacement.
- **Latency.** The serving-time Oracle is a read against persisted artefacts (`v1b_bounds.parquet` plus two CSV surfaces); byte-for-byte verified in Rust; nothing in the coverage-inversion mechanism requires a live heavy-weight forecast computation. Performance budgeting is a §8 concern, not a correctness one.

These limitations are disclosures, not retractions. Each has a concrete follow-up in §10.

10 Future Work

The limitations of §9 are disclosures, not retractions. This section names concrete follow-ups in five categories: methodology upgrades that wait on data accumulation (§10.1), methodology upgrades that wait on engineering effort (§10.2), empirical-comparator and live-deployment work (§10.3), the two companion papers that complete the trilogy (§10.4), and scope-expansion paths beyond US-equity weekends (§10.5). The full operational roadmap with quarter-grain triggers lives in `docs/v2.md` and `docs/ROADMAP.md`; this section reproduces the items that are paper-relevant and explains the precondition each one waits on.

The organising principle is simple: the present draft already demonstrates that a calibration-transparent interval oracle is useful, but the same diagnostics that keep the paper honest also define the next research frontier. Berkowitz rejection marks the boundary between per-anchor calibration and full-distribution calibration. DQ rejection marks the boundary between simple violation de-clustering and a richer conditional-dynamics model. The remaining items below should therefore be read as a deliberately gap-driven V2 agenda rather than as an unprioritised wishlist.

10.1 Data-gated methodology upgrades

Four upgrades are blocked only on data history. Each becomes runnable as soon as its trigger fires.

V2.1 — F_tok forecaster (on-chain xStock TWAP). The most likely reviewer ask, and the most-watched §9 item. Cong et al. show that on-chain tokenised-equity prices anticipate Monday opens during closed-market intervals; our base forecaster does not consume this signal because the Solana tokenised-equity launch (mid-2025) leaves only ~30 weekends of post-launch on-chain history through the cutoff of this paper, well short of the ~150 weekends needed for stable per-(symbol, regime) Kupiec validation. The V5 forward-cursor tape (`docs/v5-tape.md`, continuous capture since 2026-04-24) is the data source. ETA is Q3–Q4 2026 at current accumulation. The deliverable is a third forecaster `F_tok` and a re-derivation of the §6 OOS results admitting it into the candidate pool $\mathcal{F}$ alongside `F1_emp_regime` and `F0_stale`.

V2.2 — Rolling calibration-surface rebuild. The deployed surface is a one-shot fit on pre-2023 data. A production deployment will see the buffer schedule drift as the calibration target ages relative to the live regime. The fix is a monthly cron that rebuilds `v1b_bounds.parquet` and the two surface CSVs, re-measures `BUFFER_BY_TARGET` from each rebuild's held-out slice, and writes the fresh values to `buffer.json` for hot-reload. The trigger is the live-deployment window (§9.7); until then, the static surface is correct. Paper-side, the deliverable is a §6 buffer-trajectory figure across 6 monthly rebuilds.

V2.3 — MEV-aware consumer-experienced coverage. §9.12's open question. Requires Solana mempool-or-bundle granularity to reconstruct the price a consumer would have *executed at* near band-edge events, not just the mid we measure against P_t. The V5 tape supplies on-chain prints; Jito-bundle reconstruction supplies the ordering. ETA is Q3 2026 once Jito bundle data accumulates 3 months. The deliverable is a §9 disclosure of the gap $\Delta =$ reported $-$ consumer-experienced in basis points, not a methodology change.

V2.4 — Distribution-level calibration upgrade. §6.4.1 reports Berkowitz on continuous PITs reconstructed from a 19-point grid; the test rejects, and we attribute the rejection to buffer-induced safe-direction over-coverage outside $[0.68, 0.99]$. The immediate follow-up is diagnostic — full distributional KS plus a per-regime Berkowitz decomposition — but the real V2 target is methodological: move from an anchor-calibrated interval oracle toward a more continuous distributional correction. Candidate paths are a denser claimed-quantile grid, a direct CDF-calibration layer, or a continuous correction replacing the current flat-extrapolated anchor schedule. The deliverable is not just "more tests"; it is a clearer answer to whether Soothsayer can graduate from per-anchor calibration to distribution-level calibration without sacrificing the current deployment-facing simplicity.

10.2 Engineering-gated methodology upgrades

Three upgrades require coding effort, not data accumulation.

Conformal / continuous-calibration upgrade. §9.4's principal weakness is that the per-target buffer is a heuristic, not a theorem. We tested vanilla split-conformal (Vovk), Barber et al. (2022) nexCP at 6/12-month half-lives, and a block-recency baseline; all three under-corrected the OOS gap relative to the per-target heuristic on this data, with bootstrap CIs that exclude zero (`reports/v1b_conformal_comparison.md`). Conformal is therefore *not* an obvious win on the existing claimed-coverage grid + sample size — but two changes flip the comparison: a finer claimed-quantile grid above 0.995 (already partially landed in the 2026-04-25 grid extension to {0.997, 0.999}) shifts the inversion regime where conformal corrections are operationally meaningful, and a multi-split walk-forward evaluation distinguishes drift from sampling noise on the buffer comparison itself. The conformalised quantile-regression variant of Romano-Patterson-Candès (2019) is the closest methodological match: the F1 log-log volatility regression becomes a quantile regressor, and the empirical buffer becomes the conformal correction. More broadly, this work package asks whether the current piecewise anchor schedule should survive at all, or whether a continuous correction fit can dominate it on both interpretability and out-of-sample stability.

Sub-regime granularity and richer state. The §6.4.1 DQ-test rejection at all four anchor flags multi-lag conditional structure that Christoffersen's two-state Markov independence test misses. The probable mechanism is volatility-regime persistence within `normal` (post-shock weekends, calm range-bound weekends, recovery weekends) that the current three-state ρ aggregates. Splitting `normal` into sub-regimes — using either trailing realised-vol tertiles or a rolling

regime classifier — should absorb part of the multi-lag autocorrelation. A second candidate mechanism is omitted state: on-chain tokenized-stock prices, weekend futures microstructure, or event-level covariates may carry information the current regime labeler discards. The DQ rejection is therefore not just a call for more buckets; it is evidence that the current state representation still leaves predictable structure in misses, which is exactly the gap a richer Solana-native tape can help close.

Asymmetric / Student-t F0. F0 is a Gaussian fallback. A Student-t or skew-normal alternative would reduce the high-vol band over-coverage noted in §6.3 in-sample. Mechanical port; deliverable when re-evaluating `REGIME_FORECASTER` becomes part of a v2 surface rebuild.

10.3 Empirical comparators and live deployment

Two paper-relevant items with no methodology component.

Public comparator dashboard (closes §9.8). A side-by-side numerical comparison against incumbent oracles, on a matched evaluation window. Three of the four comparators have partial in-repo infrastructure: Pyth Hermes scrape (`reports/v1b_pyth_comparison.md`, 265-obs subset; underlier-only), Chainlink Data Streams v10 reconstruction (`reports/v1b_chainlink_comparison.md`, 87-obs sample on the pre-decoder-fix dataset — needs re-derivation under the corrected v10 + v11 decoders; see §9.8 caveat), and the RedStone REST forward-cursor tape that began continuous capture on 2026-04-26 (gateway retention is 30 days, so only forward-collected weekends are reportable; underlier-only as discussed in §9.8.1). The dashboard becomes paper-grade once each of those datasets crosses the per-comparator sample-size threshold. A Phase 2 deliverable per `docs/ROADMAP.md`.

Live-deployment OOS validation (closes §9.7). The §6 OOS coverage claim is backtested. The Rust + Anchor serving stack of §8 is functionally complete and verified (75/75 byte-for-byte parity), but has not run as a production feed consumed by a real protocol. A live window — six months of devnet-then-mainnet operation against a real consumer — would expose any failure mode that requires production-environment realism (rate-limit edges, signer-rotation events, mempool-ordering near band edges). The output is a one-page §6.5 update reporting realised coverage on the live window against the same Kupiec + Christoffersen + DQ + Berkowitz battery used here. Importantly, this closes the external-validity gap; it does not, by itself, resolve the current Berkowitz / DQ rejections, which is why the statistical upgrades above remain distinct work packages rather than presumed consequences of going live.

10.4 Companion papers (trilogy)

The §3.5 non-claims and the §9.9 disclosure (no optimal-liquidation-policy benchmark) signal that two independent decision-theoretic questions are out of scope for the coverage-inversion primitive paper. Both are addressed by companion papers already in plan.

Paper 2 — OEV mechanism design. Given a calibration-transparent oracle, what auction or trigger mechanism for oracle-extractable value (OEV) maximises protocol welfare? The existing OEV-recapture literature (Chainlink SVR, RedStone Atom, API3 OEV, UMA Oval, Pyth Express Relay) is written under an opaque-oracle assumption that calibration-transparent oracles explicitly violate; the calibration receipt of §4.5 is a primitive the auction can directly condition on. Deliverable in plan: an auction simulator (M0–M4 instantiation with builder/searcher coalition modelling), a theoretical equilibrium result on rent monotonicity in band sharpness, and an empirical-replay panel on the historical Aave V2/V3 + Kamino liquidation panel. See `reports/paper2_oev_mechanism_design/plan.md`.

Paper 3 — Optimal liquidation-policy defaults. Given a calibrated band, what should a lending protocol do? The decision-theoretic mapping from band $[L_t, U_t]$ to action $\{\texttt{Safe}, \texttt{Caution}, \texttt{Liquidate}\}$ requires the three ingredients §9.9 enumerates: a borrower-book LTV weight distribution, a protocol-specific cost model, and a declared semantics for what counts as the correct realised action. The existing protocol-compare scaffolding (`scripts/run_protocol_compare.py`, `src/soothsayer/backtest/protocol_compare.py`) plus the planned walk-forward extension supports this paper. See `reports/paper3_liquidation_policy/plan.md`.

The trilogy structure — methodology (this paper) → mechanism (Paper 2) → policy (Paper 3) — is explicitly designed so that each layer's deliverable is independently citable. A reviewer of the present paper need not commit to the trilogy framing; the coverage-inversion primitive stands alone as an oracle contract regardless of how downstream consumers use it.

10.5 Scope expansion

Multi-region replication (closes §9.10). All ten tickers in this evaluation are US-listed; the factor switchboard privileges US-session futures. A natural replication target is tokenised LSE equities (factor `FTSE`, vol index `^VFTSE`)

and tokenised JP equities (factor `^N225` or `NK=F`, vol index `^VXJ`). The methodology generalises by construction (the `FACTOR_BY_SYMBOL` and `VOL_INDEX_BY_SYMBOL` switchboards of §5.4 are designed for this); the deliverable is empirical evidence that the surface achieves §6-comparable Kupiec / Christoffersen pass rates on a non-US weekend window.

Additional RWA classes — the methodology as a missing-layer for the broader RWA wave. GLD and TLT in the present panel exercise non-equity factor + vol-index pairs and serve as anchors that the calibration-transparency primitive is not US-equity-specific. The tokenized RWA wave currently moving on-chain is materially broader than the equity slice this paper evaluates: tokenized treasuries (BlackRock BUIDL and Ondo OUSG together exceed $5B AUM as of submission), tokenized commodities (Paxos PAXG, with silver and broader baskets in flight), tokenized credit and FX derivatives, and the equities-on-Solana xStock universe that motivated the present evaluation. Each new class is, methodologically, a new `(factor, vol_index)` pair plus a re-fit of the calibration surface — no methodology change is required.

What changes across these classes is the *information set* available during the closed-market window. Tokenized US equities have ES/NQ futures + VIX. Tokenized gold has GC futures + GVZ. Tokenized treasuries have ZN/ZF/ZB futures + MOVE. Tokenized credit has CDS spreads + index volatility. Each pair satisfies the methodology's requirement of a continuous off-hours information set; for any such class, the calibration-transparent primitive is directly applicable and the public-good infrastructure it provides — auditable bands at consumer-specified τ, with on-chain receipts — generalises by construction.

§9.10 enumerates the explicit out-of-scope cases: real estate (no off-hours information set; NAV updates are discrete and infrequent), illiquid agricultural commodities (futures exist but are discontinuous), and instruments whose primary discovery happens at irregular cadences. We do *not* claim every RWA needs Soothsayer-shaped calibration infrastructure — only that for the substantial subset with continuous off-hours data, it is the missing infrastructure layer that responsible institutional-scale onboarding requires. `docs/methodology_scope.md` in the artifact accompanying this paper carries the per-class fit/no-fit table.

The deliverable for this future-work item is therefore not "extend Soothsayer to one new ticker"; it is "demonstrate the calibration-transparency primitive on a representative sample from each broad RWA class with a continuous off-hours information set, using the same Kupiec / Christoffersen disclosure framework as §6, on a panel that protocol integrators and institutional risk teams already speak."

The full operational roadmap — including quarter-grain ETA estimates, dependencies, and items not directly tied to a paper deliverable — lives in `docs/ROADMAP.md` and the methodology evolution log at `reports/methodology_history.md`. The present section enumerates only those follow-ups that close a §9 limitation, sharpen a §6 result, or anchor a planned companion paper.

11 Conclusion

This paper specified a calibration-transparent oracle primitive for tokenised real-world assets, evaluated it on 12 years of US-listed weekend data, and shipped a reference implementation with end-to-end Python Rust on-chain parity. The contribution is the *interface*, not the forecaster: any base f can be wrapped in the same surface-inversion machinery, and the consumer-facing contract is a coverage SLA rather than a point estimate.

11.1 The three properties, revisited

The contract of §3.4 holds on the deployed system as follows.

- **(P1) Auditability.** The calibration surface $S^f(s, r, q)$ is reconstructible from public data — yfinance daily prices, the per-symbol vol indices (VIX / GVZ / MOVE), and the per-symbol factor returns (ES / NQ / GC / ZN / BTC). The per-read `PricePoint` exposes the nine-field receipt of §4.5 (target coverage, calibration buffer applied, claimed coverage served, forecaster used, regime, sharpness, calibration path, bracketing pair, regime-forecaster policy snapshot), and the on-chain `PriceUpdate` PDA mirrors that receipt under the byte-for-byte parity contract of §8.5. A third party in possession of the persisted surface CSVs and the bounds parquet can reconstruct any served band from the receipt alone.
- **(P2) Conditional empirical coverage.** On the temporally disjoint 2023+ slice (1,720 rows × 172 weekends, calibration surface frozen on the 4,266 pre-2023 rows; per-anchor empirical buffer deployment-tuned on

the same slice and walk-forward-stable per §6.4), the served band at $\tau = 0.95$ realises coverage 0.950 with Kupiec $p_{uc} = 1.000$ and Christoffersen $p_{ind} = 0.485$ (§6.4). The Kupiec + Christoffersen conjunction passes at three of four anchor on the deployment grid; the = 0.99 row is a level-attribution failure with bounded magnitude (§9.1, §6.4.1). The Engle–Manganelli DQ test rejects at all four anchors including = 0.95 (§6.4.1), indicating multi-lag conditional structure beyond Christoffersen's two-state independence test; the empirical content of the P2 statement is therefore *per-anchor calibration* in the Kupiec + Christoffersen sense, not full-distribution calibration (Berkowitz also rejects, §6.4.1) and not freedom from multi-lag conditional structure (DQ rejects, §6.4.1). A 50-level inter-anchor sweep across $\tau \in [0.52, 0.98]$ passes Kupiec uniformly with $\sup |\text{realised} - \tau| < 0.025$, supporting the linear-interpolation of $b(\tau)$ across the deployment range. The number is *deployment-calibrated and walk-forward-stable* rather than purely held-out end-to-end; §10.1's V2.2 (rolling calibration-surface rebuild on a live deployment window) is the upgrade that would close the remaining gap.

- **(P3) Per-regime serving efficiency.** The hybrid policy `REGIME_FORECASTER` = {`normal` → `F1_emp_regime`, `long_weekend` → `F1_emp_regime`, `high_vol` → `F0_stale`} is the *empirically preferred deployment choice* on the OOS sample, *weakly dominating* F1-everywhere on (mean bandwidth, Christoffersen-independence margin): at the deployed buffer the hybrid simultaneously tightens bands by -2.6% versus F1-everywhere (456.0 vs 467.9 bps) and improves the Christoffersen margin from $p_{ind} = 0.221$ to $p_{ind} = 0.485$, with no observed trade-off on either axis. Both F1-everywhere and the hybrid pass Kupiec and Christoffersen at $\alpha = 0.05$ on the deployed configuration, so the hybrid is a margin-and-sharpness optimisation rather than a binary verdict-flip. The Christoffersen p-value is a violation-clustering diagnostic, not a decision-theoretic optimisation target, so the property is empirical-dominance on this sample under this evaluation battery rather than a theorem-shaped optimality claim; the reframing from in-sample bandwidth dominance to out-of-sample empirical dominance is disclosed in §9.6 and §3.4.

11.2 What the ablation says is doing the work

The §7.5 taxonomy reduces to three load-bearing components.

1. **Factor-switchboard point + empirical residual quantile.** The A0 → A1 ladder rung (factor-adjusted point + empirical-quantile bands replacing stale-Gaussian) tightens pooled bands by 39.3% (CI [37.3%, 41.3%]) at $n = 5,986$. Every subsequent ladder rung either holds sharpness flat or trades a regime-localised widening for a regime-localised coverage gain.
2. **Per-target empirical buffer.** The serving-layer C0 → C4 transition (§7.4) closes the OOS coverage gap by +2.9pp (CI [+2.0, +3.8]). Without the buffer, surface inversion alone delivers realised 0.922 at target 0.95 and Kupiec rejects. The deployed schedule $b \in \{0.68 \to 0.045, 0.85 \to 0.045, 0.95 \to 0.020, 0.99 \to 0.010\}$ is the smallest per-anchor buffer satisfying the §3.4 contract on the OOS slice.
3. **Hybrid regime-to-forecaster policy.** The *empirically preferred deployment choice* on the OOS sample for `high_vol` weekends, weakly dominating F1-everywhere on (sharpness, Christoffersen-independence margin), as discussed under (P3) above. Tightens pooled bands by 2.6% versus F1-everywhere and lifts the Christoffersen margin from $p_{ind} = 0.221$ to $p_{ind} = 0.485$, with no observed trade-off on either axis.

The remaining components (VIX-scaled standardisation, log-log vol regression, per-symbol vol index, long-weekend regressor) are either regime-localised insurance trades or generalisation anchors for the non-equity tickers; the earnings-next-week flag is disclosed for auditability with no detectable performance contribution at our sample size and is flagged for finer event-granularity replacement in v2 (§9.5).

11.3 Honest framing

Two scoping points warrant restatement at the close of the paper. First, the served band is *per-anchor calibrated*, not *full-distribution calibrated*: the Berkowitz joint LR rejects on inverse-normal-transformed PITs reconstructed from a 19-point quantile grid (§6.4.1), and the rejection is mechanically traceable to the per-target buffer schedule's flat extrapolation outside $[0.68, 0.99]$. The deviation is in the *safe direction* — over-coverage at low rather than under-coverage — and the four anchor themselves land on the diagonal of the reliability diagram. The product contract was always per-anchor; the Berkowitz disclosure makes the boundary explicit.

Second, the = 0.99 structural ceiling has bounded worst-case impact rather than open-ended tail risk. The §6.4.1 exceedance-magnitude diagnostic reports that the 40 missed-coverage events at = 0.99 have median breach 72 bps and maximum breach 796 bps, and the maximum breach shrinks monotonically across the four anchors as increases ($2,339 \to 2,150 \to 1,415 \to 796$ bps). The rejected coverage at = 0.99 is a level-attribution failure, not a tail-blowup

failure; the protocol-impact cost of that failure is bounded by ~8% on a missed event in the worst case and ~0.7% at the median. The §9.1 disclosure stands, and is now quantified.

Third, the remaining statistical gaps are useful bad news rather than project-fatal news. Berkowitz and DQ do not say that the coverage-inversion primitive is void; they say that the present implementation has reached the boundary of what an anchor-calibrated interval oracle can claim. That boundary is exactly what defines the V2 roadmap of §10: live rolling rebuilds to measure drift, richer on-chain signals and state variables to absorb conditional structure, and a more continuous calibration layer that can be evaluated as a candidate replacement for the current heuristic buffer.

11.4 What sits in the companion papers

The §3.5 non-claims and the §9.9 disclosure bracket two decision-theoretic questions that the present paper cannot answer with the evidence it presents. *Paper 2* (OEV mechanism design) addresses what auction or trigger mechanism for oracle-extractable value maximises protocol welfare given a calibration-transparent oracle — the per-read receipt of §4.5 is a primitive an auction can directly condition on, in a way the existing OEV-recapture literature does not contemplate. *Paper 3* (optimal liquidation-policy defaults) addresses the mapping from a calibrated band to a $\{\texttt{Safe}, \texttt{Caution}, \texttt{Liquidate}\}$ action under an explicit borrower-book LTV distribution and a protocol-specific cost model. The trilogy structure — methodology (this paper) → mechanism (Paper 2) → policy (Paper 3) — is designed so each layer is independently citable. A reviewer of the present paper need not commit to that framing; the coverage-inversion primitive stands alone as an oracle contract, regardless of how downstream consumers use the band.

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